

검사 협정 및 초도품 승인 서류 목록표

차종		AX	품명	ASSIST-PUMP ASSY - SUB		품번	1058-N90260-10		
제출사유		신규	부품등급	보안	EO No.	21-014	공급자명	(주)승정	
순	자료명				단계별 해당서류		첨부 서류		비 고
					검사협정	ISIR	미첨부	첨부	
1	검사협정 자료 목록표				■	■	□	■	
2	사양변경서 (변경 시)				□	■	■	□	해당사항 없음
3	초도품보증서				□	■	□	■	
4	검사협정서				■	■	□	■	
5	검사기준서				■	■	□	■	
6	외관검사기준서				■	■	□	■	
7	검사 성적서				■	■	□	■	
8	전치수 성적서(DIM'S Measurement Report)				□	■	□	■	
9	재질성적서 (MS)				□	■	□	■	
10	신뢰성시험 계획 및 결과(갑지)				■	■	■	□	해당사항 없음
11	신뢰성시험 성적서(ES 혹은 자체 시험)				□	■	□	■	
12	컨넥터 매칭성 TEST REPORT (해당 시)				□	■	■	□	해당사항 없음
13	공정관리 계획서 (필요 시 사출조건표 포함)				■	■	□	■	
14	TO 보고서 (사출 부품만 해당)				□	■	□	■	
15	공정FMEA / 설계FMEA				■	■	□	■	
16	2/3차 공급자 현황				■	■	□	■	
17	2,3차 공급업체와의 검사협정 사본				■	■	■	□	해당사항 없음
18	과거부품문제 예방대책 점검표				■	■	□	■	
19	검구/측정 보조구 LIST				■	■	□	■	
20	검구 사양서(C/F PLAN'G SHEET)				■	■	■	□	해당사항 없음
21	검구 검사 성적서				□	■	■	□	해당사항 없음
22	공정감사 평가표 (공급자 자체)				□	■	□	■	
23	공정감사 개선완료 보고서				□	■	□	■	
24	공정능력(Cpk) 결과				□	■	□	■	
25	게이지 R&R결과				□	■	□	■	
26	측정표준서				□	■	□	■	
27	한도 견본표				□	■	■	□	해당사항 없음
28	납땜품질 확인 성적서				□	■	■	□	해당사항 없음
29	전자소자 적정성 성적서				□	■	■	□	해당사항 없음
30	소프트웨어 확인 레포트				□	■	■	□	해당사항 없음
31	IMDS REPORT				□	■	□	■	
32	포장사양 승인서				□	■	□	■	
33	조립 적합성 확인서 (수요자 작성)				□	□	■	□	해당사항 없음
34	최종 EO 도면				■	■	□	■	

▶ 미첨부 서류는 비고란에 사유작성

검사 협정서 (갑지)



1.적용범위 : 이 협정은 아래 부품에 적용된다.

차 종	AX	정기검사 구분	☐ 난연성/배광 ☐ 법규(R) ☒ 보안(C) ☐ 중요,일반
		부품품질 위험도(IRE)	☒ 특별관리(고) ☐ 중요관리(중) ☐ 일반관리(저)
품 명	ASSIST-PUMP ASSY- SUB	품 번	1058-N90260-10
			☐ 해당 ☒ 해당무

2.정기검사 :

- 공급자는 검사기준서상 정기검사 해당항목을 초도품 승인시 실시한 품질확인과 동일하게 검사한다.
- 검사규격,주기 및 시료 수는 본 검사협정서와 검사기준서에 따른다.
- 정기검사 결과는 문서로 보관해야 하며,수요자 제출 요구 시는 즉각 제출되어야 한다.

3.일상검사 :

검사협정 시 제출된 공정관리계획(CONTROL PLAN)에 따르고 그 결과를 문서화 하여, 수요자 제출 요구 시 즉각 통보 되어야 한다.

4.중요검사항목 :

상기 부품의 특성을 감안, 공정 및 부품의 중요 검사항목을 아래와 같이 결정하며, 그 결과는 문서화하여 수요자 제출 요구 시는 즉각 통보 되어야 한다.

순	중요검사항목	중점관리항목	시료수	주기	관리구분	비고
1	외관	SEALING 면 : 싱크 마크, 버, 파팅 라인, 게이트 마크 없을 것.	n=5	LOT	I	
2	DIM'S	검사 기준서 만족할 것	n=5	LOT	I	
3	LEAK 검사	600kPa 공기압을 넣어 BALL 압입부에 LEAK 및 제품 파손 없을 것 (10kPa 이하 관리 할 것.)	전수	LOT	C	
4	부압 검사	500kPa 가압시 -5.0±1kPa 일 것	전수	LOT	C	
5	BALL 이탈력	노즐부 절개 후 Ø2.8봉으로 200mm/min속도로 평가시 MIN 11kgf 이상일 것	n=3	LOT	I	

※ 관리구분: 특별특성 (C)M, SPC (S), 중요관리항목 (I)

5. 미국(NHTSA :National Highway Traffic Safety Administration) 안전부품의 분류는 미국교통부 조기 보고에 의한 분류 (18시스템) #첨부 참조

6. 설계 및 4M 변경 시 공급자는 수요자측에 사전통보하고 「초도품승인 HK-TQ-PJ-016」 업무표준에 의거 초도품 재 승인을 실시 하여야 한다.

7. 공급자는 본 검사협정에 준한 품질관리 활동을 충실히 수행하고 불충실 이행으로 품질문제 발생시는 모든 책임을 진다.

8. 기타

					협정 체결일	공급자	㈜승정	
					2021년02월08일		팀(부서)명	개발품질팀
							품질보증	부 장
							책임자	허 진석
					시 행 일	수요자	현담산업㈜	
					21년02월08일 부터 "B/OUT"까지		팀(부서)명	신차품질실
							팀장 (부서장)	부 장
								박 동희

검사 협정서 (을지)



순	중요검사항목		중점관리항목	시료수	주기	관리구분	비고
1	외 관		제품에 유해한 흠집, BURR, 벌어짐 등이 없을 것	n=5	LOT	I	
			조립&TEST 후 이물부착, 미조립, 깨짐 등이 없을 것				
			SEALING면 싱크 마크, BURR, 파팅 라인, 게이트 마크 없을 것				
			부품의 누락이 없을 것				
2	재 질	BODY	KEP F25-03/ KEP F20-03,KOLON K500 Polyplastics M 90-44 또는 상당 재질일 것. H/KMC POM-009 TYPE A(MS237-09)따를 것.	n=1	1회/반기	I	HIC 사급
		BALL	SUS304-WPB				
		O-RING	FKM				직거래
3	DIM'S	①	Ø25.5 0/-0.1	n=5	LOT	I	
		②	Ø20 +0.2/0				
		③	Ø23.0 +0.2/-0.1				
		④	8.0 ±0.1mm				
		⑤	2 +0.2/0mm				
		⑥	6 +0.2/0mm				
		⑦	Ø7.9 ±0.1				
		⑧	Ø3.3±0.1				
		⑨	Ø4.18 0/-0.2				
		⑩	6 ±0.1mm				
		⑪	Ø5 +0.2/0				
		⑫	Ø0.48 ±0.05				
		⑬	◎동심도 0.2이하				
		⑭	Rz3.2 이하				
4	중 금 속		MS201-02 만족할 것	n=1	1회/반기	I	
5	LEAK 검사		600kPa 공기압을 넣어 BALL 압입부에 LEAK 및 제품 파손 없을 것 (10kPa 이하 관리 할 것.)	전수	LOT	C	
6	부압 검사		500kPa 가압시 -5.0±1kPa 일 것	전수	LOT	C	
7	BALL 이탈력		노즐부 절개 후 Ø2.8봉으로 200mm/min 속도로 평가시 MIN 11kgf 이상일 것	n=3	LOT	I	
8	중량		BODY : 5.1±0.3g	n=1	LOT	I	
			ASS'Y : 5.8±0.3g				
9	관통부 막힘 검사		모든 유로부 막힘 및 BURR 없을 것.	전수	LOT	I	

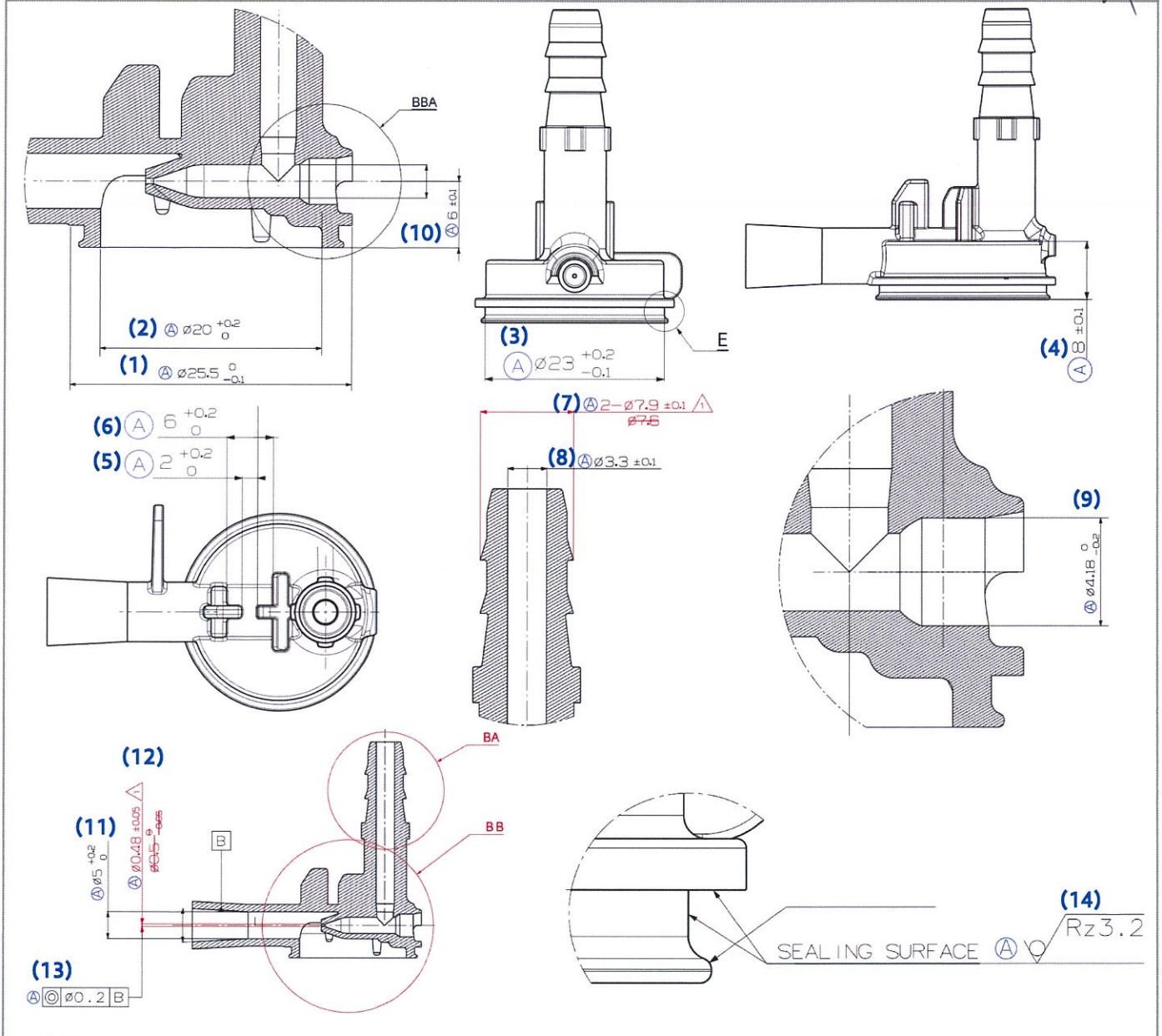
※ 관리구분: 특별특성 (C), SPC (S), 중요관리항목 (I) : 실시 대상 항목을 표기.

검사 기준서 (갑지)



작성일 : 2021년 07월 08일

차종	AX	품 명	ASSIS- PUMP ASSY- SUB	품 번	1058-N90260-10	공급자	㈜승 정
부품중요도		☐ 보안 ☐ 법규 ☐ 중요 ☒ 일반				품질 보증 책임자	
부품품질위험도(IRE)		☐ 특별관리(고) ☐ 중요관리(중) ☒ 일반관리(저)				허 진석 부장	



NO	EO No (EO 일자)	개 정 사유	공급자			수요자	
			품질			설계	확인
			작성	검토	승인		
0	21-014(21.02.08)	PILOT RELEASED	김영	✓	박	조	

※ 승인 : 공급자(품질보증 책임자,설계팀장), 수요자(초도품 승인부문 보직과장 또는 W/GROUP장)

검사 기준서 (을지)



작성일 : 2021년 07월 08일

순위	검 사 항 목		측정 GAUGE	규 격	관리 구분	시료수	검사주기		비 고
							공급자	수요자	
1	외 관		육 안	제품에 유해한 흠집, BURR, 벌어짐 등이 없을 것	I	n=5	LOT		
				조립&TEST 후 이물부착, 미조립, 깨짐 등이 없을 것					
				SEALING면 싱크 마크, BURR, 파팅 라인, 게이트 마크 없을 것					
				부품의 누락이 없을 것					
2	재 질	BODY	MILL SHEET	KEP F25-03/ KEP F20-03,KOLON K500 Polyplastics M 90-44 또는 상당 재질일 것. H/KMC POM-009 TYPE A(MS237-09) 따를 것	I	n=1	1회 /반기		HIC 사급
		BALL		SUS304-WPB					직거래
		O-RING		FKM					
3	중 금 속		중금속 성적서	MS201-02를 만족할 것	I	n=1	1회 /반기		
4	LEAK 검사		LEAK& 부압 검사기	600kPa 공기압을 넣어 BALL 압입부에 LEAK 및 제품 파손 없을 것. (10kPa 이하 관리 할 것.)	C	전수	LOT		
5	부압 검사			500kPa 가압시 -5.0±1kPa 일것	C	전수	LOT		
6	BALL 이탈력		인장검사기	노즐부 절개 후 Ø2.8봉으로 200mm/min 속도로 평가시 MIN 11kgf 이상일 것	I	n=3	LOT		
7	중 량		전자 저울	BODY : 5.1±0.3g	I	n=1	LOT		
				ASSY : 5.8±0.3g					
8	관통부 막힘 검사		육안	모든 유로부 막힘 및 BURR 없을 것.	I	전수	LOT		

※ 관리구분: 특별특성 (C), SPC (S), 중요관리항목 (I) : 실시 대상 항목을 표기.

※ 관리구분: 특별특성 (C), SPC (S), 중요관리항목 (I) : 실시 대상 항목을 표기.

DOZ
Drive Defects to Zero

작성일 : 2021년 07월 08일

[illegible]

※ 관리구분: 특별특성 (C), SPC (S), 중요관리항목 (I) : 실시 대상 항목을 표기.

외관검사기준서

결재

작성
김영

검토

승인

신

차종

품번

품명

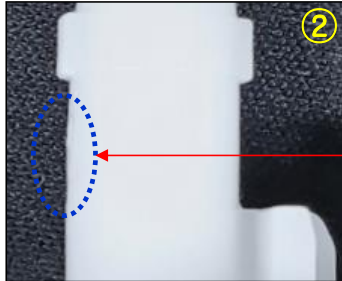
고객명

AX

1058-N90260-10

ASSIST-PUMP ASSY- SUB

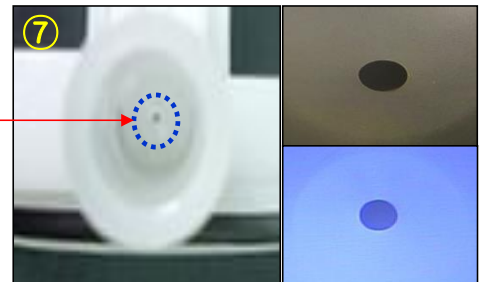
현당산업 (주)



돌출, 크랙, BURR 없을 것



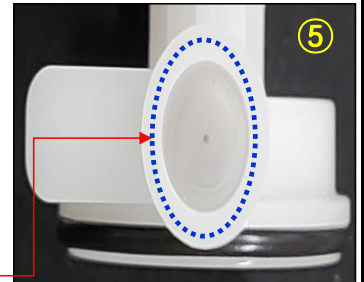
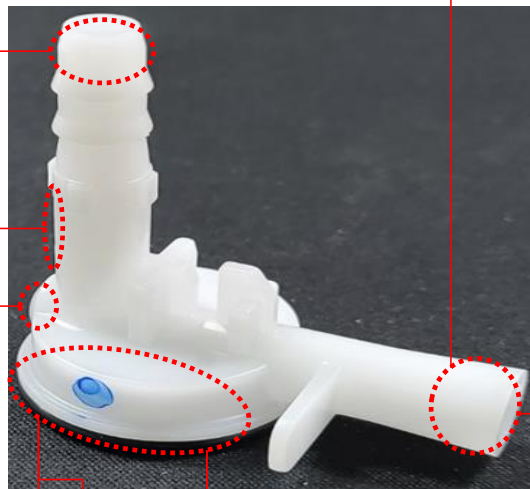
BURR, 찍힘, 미성형, 파팅
BURR, 돌출 없을 것



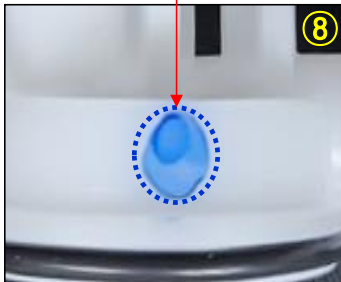
노즐부 미성형, BURR, 막힘 없을 것



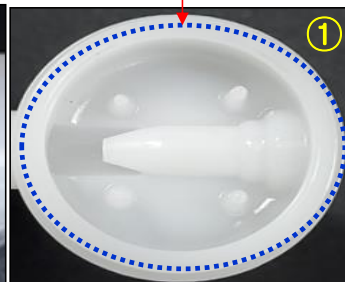
BALL 누락, 미압입, BURR, 이물, 크랙,
찍힘 없을 것,
BALL 압입시 이중 압입 유무 확인 할
것.



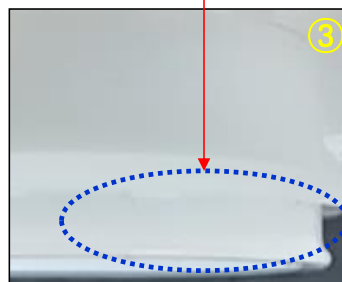
BURR, 미성형, 크랙, 편심, 찍힘
없을 것



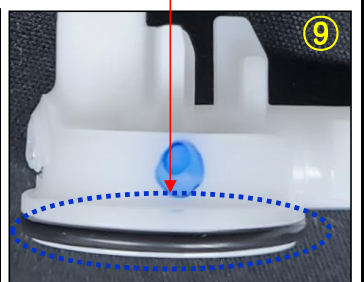
마킹 위치, 크기 및 누락 이상
없을 것



미성형, 찍힘, BURR, 벌어짐,
크랙 백화 등이 없을 것



싱크 마크, BURR, 피팅 라인,
게이트 마크 없을 것



O-RING 누락 없을 것

NO	관리 항목	관리 규격	과거차 반영	검사구	주기
1	CUP 조립부	미성형, 찍힘, BURR, 벌어짐, 크랙, 백화 등이 없을 것	X	육안	n=5
2	GATE 부	돌출, 크랙, BURR 없을 것	O	육안	n=5
3	SEALING면	싱크 마크, BURR, 피팅 라인, 게이트 마크 없을 것	X	육안	n=5
4	HOSE 조립부(포트리부)	BURR, 찍힘, 미성형, 파팅 BURR, 돌출 없을 것	X	육안	n=5
5	토출부	BURR, 미성형, 크랙, 편심, 찍힘 없을 것	X	육안	n=5
6	BALL 압입 상태	BALL 누락, 미압입, BURR, 이물, 크랙, 찍힘 없을 것 BALL 압입시 이중 압입 유무 확인 할 것	O	육안	n=5
7	노즐부	미성형, BURR, 홀 막힘 없을 것	X	확대경	n=5
8	마킹 상태	마킹 위치, 크기 및 누락 이상 없을 것	X	육안	n=5
9	O-RING 조립 상태	O-RING 누락 없을 것	X	육안	n=5

제·개정 이력	순	개정일	개정 내용	승인	순	개정 내용	승인
	0	21.03.02	초도 작성				

검 사 성 적 서 (갑지)



작성일: 2021년 07월 02일

승인항목		☒ 외관 ☒ 치수 ☒ 재질(MS) ☐ 성능(ES)					종합판정		☒ 승인 ☐ 기각 ☐ 기타 ()											
차 중		AX		공급자명		㈜승 정		EO NO		21-014		공급자	작 성		검 토		승 인		수요자	확 인
품 번		1058-N90260-10		품 명		ASSIST-PUMP ASSY- SUB							감		/		3A			
검사목적		☐ P1 ☐ P2 ☒ ISIR ☐ 정기 ☐ 기타()		발행번호																
순번	검 사 항 목		규 격	관리 구분	측 정 치										CAVITY수 : 2 EA			판정		
					X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	X ₈	X ₉	X ₁₀	$\bar{X}$	R	Cp		확인	
1	외 관		제품에 유해한 흠집, BURR, 벌어짐 등이 없을 것	I	OK	OK	OK	OK	OK											합격
			조립&TEST 후 이물부착, 미조립, 깨짐 등이 없을 것	I	OK	OK	OK	OK	OK											합격
			SEALING면 싱크 마크, BURR, 피팅 라인, 게이트 마크 없을 것	I	OK	OK	OK	OK	OK											합격
			부품의 누락이 없을 것	I	OK	OK	OK	OK	OK											합격
2	재 질	BODY	KEP F25-03 재질일 것 POM-009 TYPE A(MS237-09)	I	해당무														-	
		BALL	SUS304-WPB	I	해당무														-	
		O-RING	FKM	I	재질성적서 참조														-	
3	중 금 속		MS201-02를 만족할 것	I	해당무														-	
4	LEAK 검사		600kPa 공기압을 넣어 BALL 압입부 LEAK 및 제품 파손 없을 것 (10kPa 이하 관리 할 것.)	C	0	0	1	0	1										합격	
5	부압 검사		500kPa 가압시 -5.0±1.0kPa 일것	C	-4.6	-4.7	-4.7	-4.6	-4.6										합격	
6	BALL 이탈력		노즐부 절개 후 Ø2.8봉으로 200mm/min 속도로 평가시 MIN 11kgf	I	28.14	27.08	26.66												합격	
7	중 량		BODY : 5.1±0.3g	I	5.11	5.10	5.10	5.11	5.10										합격	
			ASSY : 5.8±0.3g	I	5.77	5.78	5.77	5.77	5.78										합격	
8	관통부 막힘 검사		모든 유로부 막힘 및 BURR 없을 것.	I	OK	OK	OK	OK	OK										합격	

비 고

※ 관리구분: 특별특성(C), SPC항목(S), 중요관리항목(I).

※확인란은 공급자가 제출한 성적서를 수요자가 검증이 필요한 항목에 대해 확인후 결과를 기록하는 곳임.(실 측정치)

검 사 성 적 서 (치수)



작성일: 2021년 07월 02일

순 번	검사 부위 (Sketch)	부위	규격	관리 구분	측 정 치(Cavity #1)										CAVITY수 : 2 EA				판정
					X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	X ₈	X ₉	X ₁₀	X	R	Cp	확인	
9	DIM'S (1) $\varnothing 25.5 \begin{smallmatrix} 0 \\ -0.1 \end{smallmatrix}$ (2) $\varnothing 20 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$ (3) $\varnothing 23 \begin{smallmatrix} +0.2 \\ -0.1 \end{smallmatrix}$ (4) 8.0 ± 0.1 (5) $2 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$ (6) $6 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$ (7) $\varnothing 7.9 \pm 0.1$ (8) $\varnothing 3.3 \pm 0.1$ (9) $\varnothing 4.18 \begin{smallmatrix} 0 \\ -0.2 \end{smallmatrix}$ (10) 6 ± 0.1 (11) $\varnothing 5 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$ (12) $\varnothing 0.48 \pm 0.05$ (13) ◎동심도 0.20이하 (14) Rz3.2 이하	(1) $\varnothing 25.5 \begin{smallmatrix} 0 \\ -0.1 \end{smallmatrix}$		25.43	25.43	25.42	25.42	25.43							25.43	0.01	3.04	OK	합격
		(2) $\varnothing 20 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$		20.07	20.07	20.09	20.08	20.08							20.08	0.01	3.98	OK	합격
		(3) $\varnothing 23.0 \begin{smallmatrix} +0.2 \\ -0.1 \end{smallmatrix}$		23.11	23.10	23.12	23.11	23.10							23.11	0.01	5.98	OK	합격
		(4) 8.0 ± 0.1		7.94	7.94	7.95	7.94	7.95							7.94	0.01	6.09	OK	합격
		(5) $2 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$		2.17	2.18	2.16	2.17	2.18							2.17	0.01	3.98	OK	합격
		(6) $6 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$		6.15	6.15	6.16	6.15	6.16							6.15	0.01	6.09	OK	합격
		(7) $\varnothing 7.9 \pm 0.1$		7.96	7.95	7.96	7.94	7.96							7.95	0.01	3.73	OK	합격
		(8) $\varnothing 3.3 \pm 0.1$		3.31	3.32	3.31	3.31	3.32							3.31	0.01	6.09	OK	합격
		(9) $\varnothing 4.18 \begin{smallmatrix} 0 \\ -0.2 \end{smallmatrix}$		4.117	4.117	4.116	4.119	4.118							4.12	0.00	29.24	OK	합격
		(10) 6 ± 0.1		5.933	5.931	5.936	5.932	5.935							5.93	0.00	16.07	OK	합격
		(11) $\varnothing 5 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$		5.112	5.113	5.112	5.111	5.110							5.11	0.00	29.24	OK	합격
		(12) $\varnothing 0.48 \pm 0.05$		0.472	0.471	0.471	0.472	0.473							0.47	0.00	19.92	OK	합격
		(13) ◎동심도 0.20이하		0.130	0.121	0.131	0.114	0.128							0.12	0.01	-	OK	합격
		(14) Rz3.2 이하		0.742	0.789	0.763	0.752	0.776							0.76	0.02	-	OK	합격

(6) $\varnothing 5 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$

(5) $\varnothing 2 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$

(7) $\varnothing 2 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$

(8) $\varnothing 3.3 \pm 0.1$

(9) $\varnothing 4.18 \begin{smallmatrix} 0 \\ -0.2 \end{smallmatrix}$

(11) $\varnothing 5 \begin{smallmatrix} +0.2 \\ 0 \end{smallmatrix}$

(12) $\varnothing 0.48 \pm 0.05$

(13) ◎동심도 0.20이하

(14) Rz3.2 이하

비 고

※ 관리구분: 특별특성(C), SPC항목(S), 중요관리항목(I).

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검 사 성 적 서 (치수)



작성일: 2021년 07월 02일

순 번	검사 부위 (Sketch)	부위	규격	관리 구분	측 정 치(Cavity #2)										CAVITY수 : 2 EA				판정
					X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	X ₈	X ₉	X ₁₀	X	R	Cp	확인	
9		①	Ø25.5 0/-0.1		25.42	25.43	25.43	25.43	25.41						25.42	0.01	1.86	OK	합격
		②	Ø20 +0.2/0		20.07	20.09	20.08	20.08	20.09						20.08	0.01	3.98	OK	합격
		③	Ø23.0 +0.2/-0.1		23.11	23.10	23.12	23.10	23.11						23.11	0.01	5.98	OK	합격
		④	8.0 ±0.1		7.95	7.95	7.94	7.96	7.94						7.95	0.01	3.98	OK	합격
		⑤	2 +0.2/0		2.15	2.16	2.15	2.15	2.16						2.15	0.01	6.09	OK	합격
		⑥	6 +0.2/0		6.16	6.16	6.17	6.16	6.16						6.16	0.00	7.45	OK	합격
		⑦	Ø7.9 ±0.1		7.92	7.93	7.94	7.92	7.91						7.92	0.01	2.92	OK	합격
		⑧	Ø3.3±0.1		3.33	3.31	3.32	3.31	3.32						3.32	0.01	3.98	OK	합격
		⑨	Ø4.18 0/-0.2		4.136	4.132	4.130	4.136	4.135						4.13	0.00	12.42	OK	합격
		⑩	6 ±0.1		5.911	5.916	5.918	5.913	5.916						5.91	0.00	12.01	OK	합격
		⑪	Ø5 +0.2/0		5.126	5.128	5.129	5.122	5.131						5.13	0.00	9.75	OK	합격
		⑫	Ø0.48 ±0.05		0.475	0.471	0.474	0.472	0.473						0.47	0.00	10.54	OK	합격
		⑬	◎동심도 0.20이하		0.141	0.160	0.158	0.152	0.149						0.15	0.01	-	OK	합격
		⑭	Rz3.2 이하		0.758	0.776	0.792	0.733	0.725						0.76	0.03	-	OK	합격

비 고

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DIMENSION MEASUREMENT REPORT(A)

Supplier

Part Name	ASSIST-PUMP ASSY-SUB
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Car Model

AX

회사명

(주)승정

Part No.	1058-N90260-10
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EO NO.

21-014

승인

27

Inspection Standard

검토

✓

검사

$$\begin{array}{r} 7102 \\ 0110 \end{array}$$

TRIAL LEVEL

CHK Date

E.0 NO

21-014

Pass

 X_1

100

 x_2

Rate

 λ_3

(%)

 x_4

(10)

$$\frac{A_4}{V}$$

Average

Judgement

[illegible][illegible][illegible]

PILOT RELEASED

REV I S I O N S

Prepare	
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Checked

Approved

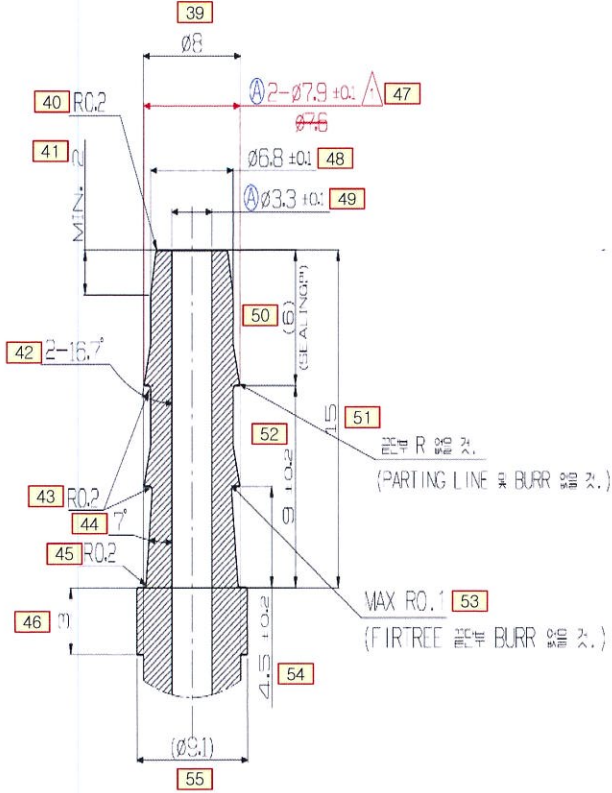
DIMENSION MEASUREMENT REPORT(B) Cavity #1

[illegible]

DIMENSION MEASUREMENT REPORT(B) Cavity #1

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance		Control Code	DATA SHEET													
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Compensation	USL		LSL	Date : 2021 / 07 / 02	X ₁	X ₂	X ₃	X ₄	X ₅				$\bar{X}$	R	Cp	판정
Sketch for inspection				20	3.8	±1	4.80	2.80		3.441	3.565	3.493	3.511	3.500				3.50	0.04	7.52	OK
				21	Ø2	±0.2	2.20	1.80		1.804	1.803	1.809	1.801	1.794				1.80	0.01	12.23	OK
				22	1.5	±0.2	1.70	1.30		1.393	1.425	1.403	1.411	1.405				1.41	0.01	5.66	OK
				23	6	-0.1	6.00	5.90		5.945	5.939	5.956	5.951	5.953				5.95	0.01	2.45	OK
				24	R3.1	±0.5	3.60	2.60		3.083	3.041	3.071	3.056	3.068				3.06	0.02	10.44	OK
				25	R4.1	±0.5	4.60	3.60		4.001	3.946	3.940	3.953	3.931				3.95	0.03	6.09	OK
				26	R0.5	±0.3	0.80	0.20		4.053	4.021	4.039	4.040	4.026				4.04	0.01	7.91	OK
				27	R1	±0.3	1.30	0.70		1.011	1.020	1.017	1.024	1.028				1.02	0.01	15.34	OK
				28	R0.5	±0.3	0.80	0.20		0.583	0.492	0.549	0.549	0.538				0.54	0.03	3.05	OK
				29	R4.1	±0.5	4.60	3.60		3.883	3.816	3.783	3.801	3.795				3.82	0.04	4.22	OK
				30	R0.5	±0.3	0.80	0.20		0.692	0.685	0.684	0.679	0.695				0.69	0.01	15.52	OK
				31	R0.5	±0.3	0.80	0.20		0.323	0.354	0.354	0.356	0.325				0.34	0.02	5.94	OK
				32	R2	±0.5	2.50	1.50		1.885	1.924	1.890	1.900	1.891				1.90	0.02	10.75	OK
				33	Ø23	+0.2/-0.1	23.20	22.90		23.067	23.071	23.072	23.064	23.049				23.06	0.01	5.38	OK
				34	9.5	±1	10.50	8.50		9.716	9.694	9.888	9.917	9.909				9.82	0.11	3.03	OK
				35	Ø2	±0.2	2.20	1.80		2.054	2.114	2.078	2.092	2.089				2.09	0.02	3.05	OK
				36	R0.5	±0.3	0.80	0.20		0.268	0.274	0.248	0.281	0.278				0.27	0.01	7.62	OK
				37	R0.5	±0.3	0.80	0.20		0.240	0.250	0.247	0.254	0.232				0.24	0.01	11.49	OK
				38	3.5	+0.1	3.6	3.5		3.534	3.539	3.534	3.521	3.544				3.53	0.01	1.95	OK
※ Control Code : Control Method Classification For the Inspection C:Special Characteristic, I:Important Inspection S:SPC(Statistical Process Control)				Pass rate =		Pass Point Total Point															

DIMENSION MEASUREMENT REPORT(B) Cavity #1

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance			Control Code	DATA SHEET													
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Compensation	Master: / /			Date : 2021 / 07 / 02													
						USL	LSL			X ₁	X ₂	X ₃	X ₄	X ₅					$\bar{X}$	R	Cp	판정
Sketch for inspection 				39	Ø8	±0.2	8.20	7.80		7.890	7.907	7.897	7.887	7.881				7.89	0.01	6.67	OK	
				40	R0.2	±0.3	0.50	0.00		0.187	0.205	0.198	0.190	0.194					0.19	0.01	11.82	OK
				41	MIN2	-2	2.00	0.00		1.929	1.955	1.914	1.935	1.904					1.93	0.02	16.94	OK
				42	16.7 °	±1	17.70	15.70		17.066	17.042	17.068	16.948	17.068					17.04	0.05	6.45	OK
				43	R0.2	±0.3	0.50	0.00		0.172	0.181	0.164	0.178	0.174					0.17	0.01	12.83	OK
				44	7 °	±1	8.00	6.00		6.411	6.399	6.514	6.410	6.428					6.43	0.05	7.13	OK
				45	R0.2	±0.3	0.50	0.00		0.178	0.192	0.191	0.188	0.183					0.19	0.01	14.23	OK
				46	3	±0.2	3.20	2.80		3.177	3.185	3.174	3.163	3.167					3.17	0.01	7.74	OK
				47	Ø7.9	±0.1	8.00	7.80		7.852	7.856	7.845	7.847	7.853					7.85	0.00	7.40	OK
				48	Ø6.8	±0.1	6.90	6.70		6.735	6.752	6.746	6.743	6.733					6.74	0.01	4.24	OK
				49	Ø3.3	±0.1	3.40	3.20		3.313	3.312	3.315	3.316	3.329					3.32	0.01	4.84	OK
				50	6	±0.2	6.20	5.80		6.177	6.177	6.183	6.189	6.160					6.18	0.01	6.16	OK
				51	15	±0.25	15.25	14.75		15.035	15.032	15.045	15.046	15.030					15.04	0.01	11.21	OK
				52	9	±0.2	9.20	8.80		8.857	8.859	8.876	8.857	8.854					8.86	0.01	7.58	OK
				53	R0.1		0.10	0.00		0.074	0.066	0.079	0.074	0.056					0.07	0.01	1.85	OK
54	4.5	±0.2	4.70	4.30		4.323	4.342	4.325	4.313	4.307					4.32	0.01	4.98	OK				
55	Ø9.1	±0.2	9.3	8.9		9.140	9.131	9.131	9.121	9.118					9.13	0.01	7.56	OK				

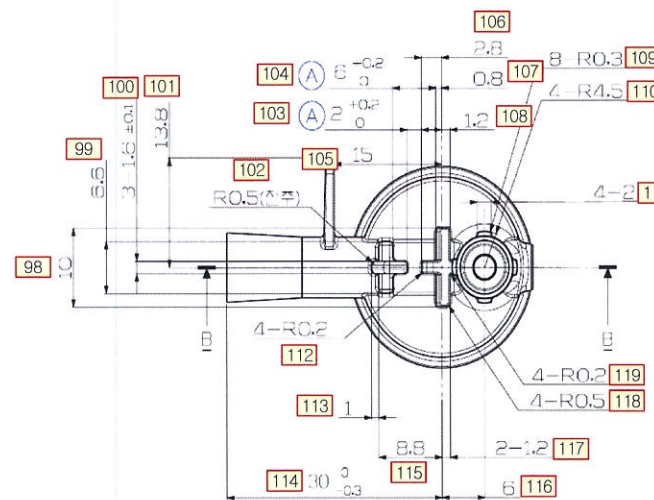
DIMENSION MEASUREMENT REPORT(B) Cavity #1

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance		Control Code	DATA SHEET														
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Master: / /			Date : 2021 / 07 / 02														
					Compensation	USL		LSL	X ₁	X ₂	X ₃	X ₄	X ₅				$\bar{X}$	R	Cp	판정		
Sketch for inspection				56	R0.2	±0.3	0.50	0.00		0.164	0.166	0.151	0.171	0.162				0.16	0.01	11.27	OK	
				57	1	±0.2	1.20	0.80		1.072	1.065	1.083	1.080	1.065				1.07	0.01	8.00	OK	
				58	1	±0.2	1.20	0.80		1.110	1.115	1.094	1.112	1.106				1.11	0.01	8.16	OK	
				59	R5	±0.5	5.50	4.50		4.894	4.911	4.924	4.891	4.896				4.90	0.01	11.94	OK	
				60	R6	±0.5	6.50	5.50		6.114	6.117	6.084	6.113	6.103				6.11	0.01	12.36	OK	
				61	Ø3	+0.1	3.10	3.00		3.025	3.013	3.011	3.021	3.008				3.02	0.01	2.34	OK	
				62	1.5x0.15	±0.2	1.70	1.30		1.474x0.140	1.480x0.141	1.469x0.151	1.499x0.145	1.481x0.147								OK
				63	4	±0.2	4.20	3.80		3.962	3.953	3.951	3.950	3.962				3.96	0.01	11.22	OK	
				64	6	±0.1	6.10	5.90		5.964	5.972	5.965	5.974	5.972				5.97	0.00	7.31	OK	
				65	Ø3	+0.1	3.10	3.00		3.055	3.050	3.048	3.060	3.050				3.05	0.00	3.42	OK	
				66	4.5	±0.1	4.60	4.40		4.543	4.537	4.541	4.541	4.539				4.54	0.00	14.62	OK	
				67	0.7	-0.1	0.70	0.60		0.649	0.651	0.649	0.646	0.667				0.65	0.01	1.99	OK	
				68	6	-0.2	6.00	5.80		5.844	5.825	5.841	5.849	5.829				5.84	0.01	3.27	OK	
				69	Ø20	+0.2	20.20	20.00		20.074	20.065	20.076	20.057	20.053				20.07	0.01	3.29	OK	
				70	Ø23	+0.2/-0.1	23.20	22.90		23.088	23.068	23.099	23.071	23.075				23.08	0.01	3.85	OK	
				71	Ø24	+0.2/-0.1	24.20	23.90		23.928	23.944	23.935	23.920	23.896				23.92	0.02	2.74	OK	
				72	Ø25.5	-0.1	25.50	25.40		25.449	25.448	25.435	25.443	25.432				25.44	0.01	2.18	OK	
				73	1.1	±0.2	1.30	0.90		1.191	1.173	1.191	1.192	1.188				1.19	0.01	8.37	OK	
				74	2	-0.1	2.00	1.90		1.911	1.915	1.908	1.918	1.922				1.91	0.01	3.01	OK	
				75	1	±0.2	1.20	0.80		1.168	1.170	1.187	1.154	1.183				1.17	0.01	5.08	OK	
				76	R0.5	±0.3	0.80	0.20		0.573	0.559	0.594	0.596	0.581				0.58	0.02	6.52	OK	
				77	R0.25	±0.3	0.55	0.00		0.226	0.216	0.242	0.230	0.216				0.23	0.01	8.44	OK	
				78	Rz3.2		3.20	0.00		0.666	0.705	0.816	0.774	0.518				0.70	0.12	4.63	OK	
* Control Code : Control Method Classification For the Inspection C:Special Characteristic, I:Important Inspection S:SPC(Statistical Process Control)				Pass rate =			Pass Point		Total Point													

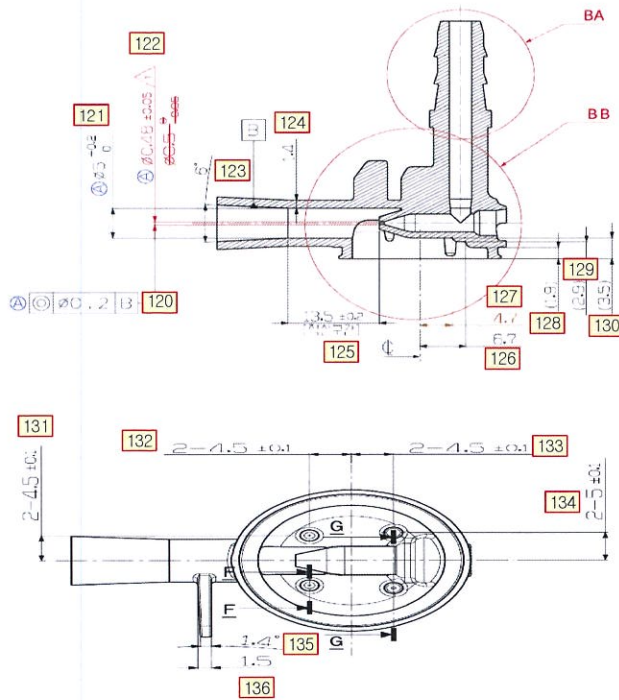
DIMENSION MEASUREMENT REPORT(B) Cavity #1

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance		Control Code	DATA SHEET													
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Master: / /			Date : 2021 / 07 / 02													
				Compensation	USL	LSL		X ₁	X ₂	X ₃	X ₄	X ₅					$\bar{X}$	R	Cp	판정	
Sketch for inspection				79	3.1	±0.2	3.30	2.90		2.942	2.933	2.937	2.947	2.934				2.94	0.01	11.38	OK
				80	R2.25	±0.5	2.75	1.75		2.104	2.044	2.069	2.080	2.003				2.06	0.04	4.33	OK
				81	R0.2	±0.3	0.50	0.00		0.273	0.251	0.274	0.268	0.266				0.27	0.01	9.02	OK
				82	R1	±0.3	1.30	0.70		1.084	1.069	1.097	1.115	1.097				1.09	0.02	5.84	OK
				83	R0.5	±0.3	0.80	0.20		0.427	0.419	0.429	0.454	0.420				0.43	0.01	7.04	OK
				84	R1	±0.3	1.30	0.70		0.991	1.006	0.998	0.985	1.009				1.00	0.01	9.97	OK
				85	R0.5	±0.3	0.80	0.20		0.434	0.481	0.440	0.431	0.445				0.45	0.02	4.95	OK
				86	2	+0.1	2.10	2.00		2.084	2.077	2.095	2.095	2.093				2.09	0.01	2.08	OK
				87	4.75	±0.2	4.95	4.55		4.644	4.655	4.651	4.671	4.650				4.65	0.01	6.55	OK
				88	2.6	+0.2	2.80	2.60		2.629	2.621	2.623	2.633	2.630				2.63	0.01	6.64	OK
				89	R1.5	±0.3	1.80	1.20		1.401	1.374	1.398	1.385	1.398				1.39	0.01	8.75	OK
				90	R0.5	±0.3	0.80	0.20		0.319	0.307	0.320	0.323	0.334				0.32	0.01	10.35	OK
				91	R1	±0.3	1.30	0.70		0.917	0.899	0.923	0.905	0.896				0.91	0.01	8.61	OK
				92	R0.3	±0.3	0.60	0.00		0.428	0.429	0.406	0.406	0.422				0.42	0.01	8.73	OK
				93	1X0.25	±0.2				0.994x0.249	1.011x0.252	0.966x0.261	0.974x0.274	0.968x0.255							OK
				94	0.25X1.5	±0.2				0.231x1.479	0.244x1.478	0.239x1.499	0.238x1.484	0.245x1.486							OK
				95	Ø3	+0.1	3.1	3.0		3.087	3.082	3.072	3.084	3.084				3.08	0.01	2.89	OK
				96	R1	±0.3	1.3	0.7		0.889	0.911	0.911	0.876	0.889				0.90	0.02	6.51	OK
				97	Ø4.18	-0.2	4.18	3.98		4.136	4.129	4.133	4.129	4.130				4.13	0.00	10.93	OK
※ Control Code : Control Method Classification For the Inspection C:Special Characteristic, I:Important Inspection S:SPC(Statistical Process Control)				Pass rate =		Pass Point Total Point															

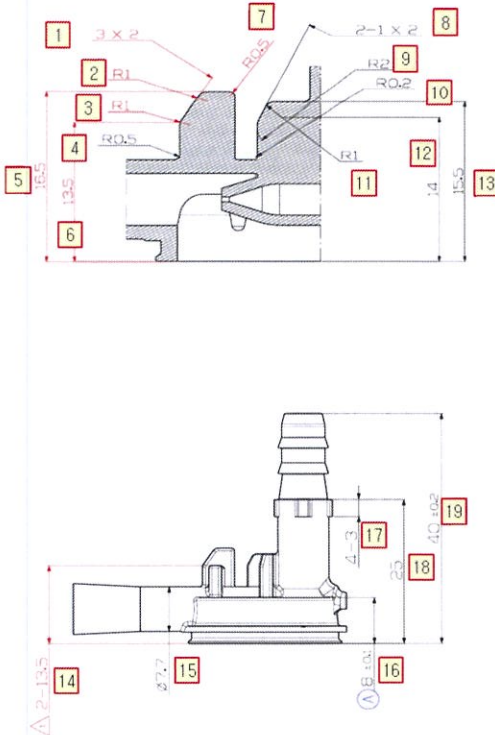
DIMENSION MEASUREMENT REPORT(B) Cavity #1

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance		Control Code	DATA SHEET													
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Compensation	USL		LSL	Date : 2021 / 07 / 02	X ₁	X ₂	X ₃	X ₄	X ₅					$\overline{X}$	R	Cp
				98	10	±0.25	10.25	9.75		10.076	10.081	10.085	10.096	10.081				10.08	0.01	11.07	OK
				99	6.6	±0.25	6.80	6.40		6.603	6.590	6.590	6.580	6.600				6.59	0.01	7.28	OK
				100	1.6	±0.1	1.70	1.50		1.691	1.684	1.697	1.696	1.691				1.69	0.01	6.45	OK
				101	13.8	±0.25	14.05	13.55		13.710	13.684	13.618	13.654	13.612				13.66	0.04	1.98	OK
				102	R0.5	±0.3	0.80	0.20		0.425	0.418	0.433	0.436	0.438				0.43	0.01	12.00	OK
				103	2	+0.2	2.20	2.00		2.161	2.165	2.159	2.157	2.145				2.16	0.01	4.42	OK
				104	6	+0.2	6.20	6.00		6.145	6.153	6.147	6.136	6.131				6.14	0.01	3.78	OK
				105	15	±0.25	15.25	14.75		15.050	15.048	15.031	15.049	15.044				15.04	0.01	10.64	OK
				106	2.8	±0.2	3.00	2.60		2.676	2.676	2.688	2.687	2.678				2.68	0.01	11.11	OK
				107	0.8	±0.2	1.00	0.60		0.668	0.657	0.665	0.669	0.677				0.67	0.01	9.23	OK
				108	1.2	±0.2	1.40	1.00		1.075	1.085	1.068	1.072	1.068				1.07	0.01	9.49	OK
				109	R0.3	±0.3	0.60	0.00		0.329	0.312	0.318	0.344	0.342				0.33	0.01	7.05	OK
				110	R4.5	±0.5	5.00	4.00		4.511	4.556	4.594	4.548	4.510				4.54	0.04	4.76	OK
				111	2	±0.2	2.20	1.80		2.100	2.110	2.100	2.110	2.120				2.11	0.01	7.97	OK
				112	R0.2	±0.3	0.50	0.00		0.208	0.196	0.214	0.202	0.198				0.20	0.01	11.26	OK
				113	1	±0.2	1.20	0.80		1.021	0.998	1.016	1.034	1.024				1.02	0.01	5.03	OK
				114	30	-0.3	30.00	29.70		29.783	29.804	29.776	29.806	29.784				29.79	0.01	3.70	OK
				115	8.8	±0.2	9.00	8.60		8.784	8.799	8.778	8.791	8.770				8.78	0.01	5.93	OK
				116	6	±0.2	6.20	5.80		6.149	6.160	6.151	6.132	6.154				6.15	0.01	6.37	OK
				117	1.2	±0.2	1.40	1.00		1.273	1.257	1.273	1.274	1.264				1.27	0.01	8.93	OK
118	R0.5	±0.3	0.80	0.20		0.437	0.433	0.452	0.440	0.423				0.44	0.01	9.47	OK				
119	R0.2	±0.3	0.50	0.00		0.196	0.184	0.203	0.190	0.212				0.20	0.01	7.61	OK				
※ Control Code : Control Method Classification For the Inspection C:Special Characteristic, I:Important Inspection S:SPC(Statistical Process Control)				Pass rate =				Pass Point		Total Point											

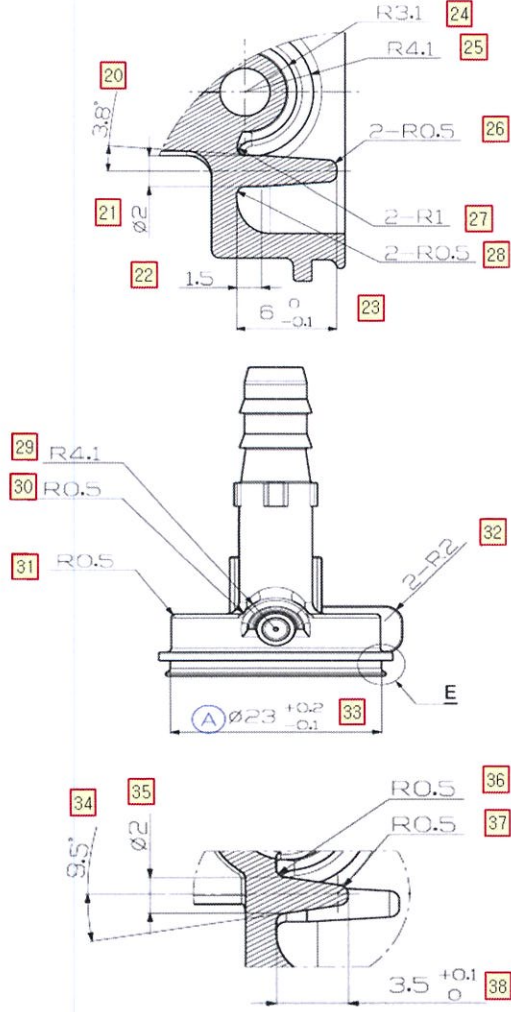
DIMENSION MEASUREMENT REPORT(B) Cavity #1

Part No.	1058-N90260-10	Inspection Area		SPEC		Tolerance		Control Code	DATA SHEET																	
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Compensation	USL	LSL		Date : 2021 / 07 / 02																	
									X ₁	X ₂	X ₃	X ₄	X ₅					$\bar{X}$	R	Cp	판정					
Sketch for inspection 									120	◎0.2		0.20	0.00		0.146	0.149	0.136	0.128	0.109				0.13	0.02	2.07	OK
									121	Ø5	+0.2	0.20	0.00		5.110	5.112	5.103	5.103	5.106				5.11	0.00	8.16	OK
									122	Ø0.48	±0.05	0.53	0.43		0.474	0.476	0.467	0.469	0.471				0.47	0.00	4.57	OK
									123	6°	±1	7.00	5.00		5.602	5.514	5.538	5.408	5.412				5.49	0.08	3.98	OK
									124	1.4	±0.2	1.60	1.20		1.262	1.258	1.249	1.258	1.269				1.26	0.01	9.18	OK
									125	13.5	±0.2	13.70	13.30		13.592	13.578	13.590	13.582	13.605				13.59	0.01	6.39	OK
									126	6.7	±0.2	6.90	3.50		6.634	6.621	6.691	6.594	6.605				6.63	0.04	14.97	OK
									127	4.7	±0.2	4.90	3.50		4.682	4.716	4.702	4.717	4.701				4.70	0.01	16.41	OK
									128	1.9	±0.2	2.10	1.70		2.045	2.058	2.053	2.058	2.056				2.05	0.01	12.27	OK
									129	2.9	±0.2	3.10	2.70		3.084	3.074	3.086	3.082	3.090				3.08	0.01	11.24	OK
									130	3.5	±0.2	3.70	3.30		3.374	3.376	3.368	3.362	3.372				3.37	0.01	12.01	OK
									131	4.5	±0.1	4.60	4.40		4.586	4.582	4.571	4.592	4.585				4.58	0.01	4.31	OK
									132	4.5	±0.1	4.60	4.40		4.454	4.465	4.476	4.432	4.441				4.45	0.02	1.88	OK
									133	4.5	±0.1	4.60	4.40		4.581	4.598	4.585	4.566	4.590				4.58	0.01	2.80	OK
									134	5	±0.1	5.10	4.90		5.031	5.023	5.025	5.054	5.021				5.03	0.01	2.47	OK
									135	1.4°	±1	2.40	0.40		1.310	1.224	1.316	1.185	1.176				1.24	0.07	4.97	OK
									136	1.5	±0.2	1.70	1.30		1.429	1.462	1.433	1.439	1.428				1.44	0.01	4.77	OK
																			</							

DIMENSION MEASUREMENT REPORT(B) Cavity #2

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance		Control Code	DATA SHEET														
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Master: / /			Date : 2021 / 07 / 02														
								Compensation	USL	LSL	X ₁	X ₂	X ₃	X ₄	X ₅				$\bar{X}$	R	Cp	판정
Sketch for inspection 				1	3X2	±0.2			3.114x2.081	3.093x2.071	3.097x2.074	3.099x2.055	3.106x2.034									OK
				2	R1	±0.3	1.30	0.70	1.063	1.060	1.046	1.057	1.038				1.05	0.01	9.55	OK		
				3	R1	±0.3	1.30	0.70	1.155	1.143	1.159	1.158	1.167				1.16	0.01	11.49	OK		
				4	R0.5	±0.3	0.80	0.20	0.374	0.375	0.388	0.376	0.388				0.38	0.01	13.98	OK		
				5	16.5	±0.25	16.75	16.35	16.433	16.413	16.432	16.426	16.420				16.42	0.01	7.93	OK		
				6	13.5	±0.25	13.75	13.25	13.510	13.505	13.521	13.504	13.511				13.51	0.01	12.33	OK		
				7	R0.5	±0.3	0.80	0.20	0.416	0.411	0.419	0.403	0.420				0.41	0.01	14.33	OK		
				8	1X2	±0.2			1.011x1.884	1.029x1.896	1.021x1.919	1.008x1.871	1.016x1.880								OK	
				9	R2	±0.5	2.50	1.50	2.326	2.345	2.313	2.309	2.321				2.32	0.01	11.84	OK		
				10	R0.2	±0.3	0.50	0.00	0.135	0.139	0.157	0.157	0.152				0.15	0.01	8.06	OK		
				11	R1	±0.3	1.30	0.70	1.028	1.044	1.026	1.038	1.039				1.04	0.01	13.02	OK		
				12	14	±0.25	14.25	13.75	13.942	13.908	13.933	13.915	13.908				13.92	0.02	5.38	OK		
				13	15.5	±0.25	15.75	15.25	15.407	15.385	15.382	15.390	15.403				15.39	0.01	7.54	OK		
				14	13.5	±0.25	13.75	13.25	13.476	13.475	13.443	13.485	13.482				13.47	0.02	4.95	OK		
				15	Ø7.7	±0.25	7.90	7.50	7.711	7.733	7.725	7.711	7.733				7.72	0.01	6.02	OK		
				16	8	±0.1	8.1	7.9	7.972	7.975	7.976	7.953	7.954				7.97	0.01	2.90	OK		
				17	3	±0.2	3.2	2.8	3.142	3.151	3.155	3.143	3.137				3.15	0.01	9.17	OK		
				18	25	±0.35	25.35	24.65	24.942	24.955	24.944	24.911	24.956				24.94	0.02	6.40	OK		
				19	40	±0.2	40.2	39.8	39.993	40.014	40.022	39.991	39.991				40.00	0.01	4.53	OK		
* Control Code : Control Method Classification For the Inspection C:Special Characteristic, I:Important Inspection S:SPC(Statistical Process Control)				Pass rate =		Pass Point Total Point																

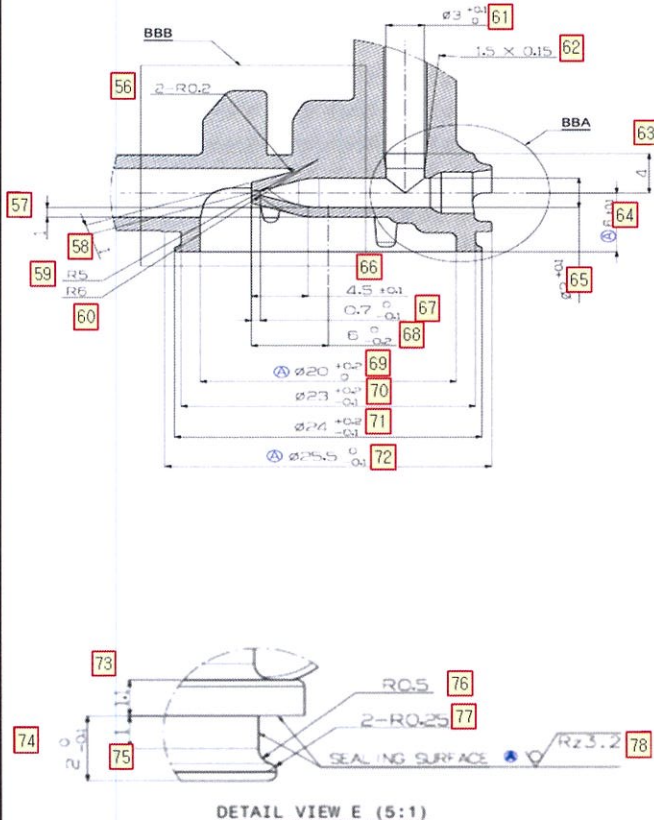
DIMENSION MEASUREMENT REPORT(B) Cavity #2

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance		Control Code	DATA SHEET														
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Master: / /			Date : 2021 / 07 / 02														
					Compensation	USL		LSL	X ₁	X ₂	X ₃	X ₄	X ₅					$\bar{X}$	R	Cp	판정	
Sketch for inspection 				20	3.8	±1	4.80	2.80		3.685	3.505	3.896	3.503	3.493				3.62	0.18	1.90	OK	
				21	Ø2	±0.2	2.20	1.80		1.824	1.805	1.833	1.826	1.803					1.82	0.01	4.97	OK
				22	1.5	±0.2	1.70	1.30		1.393	1.408	1.403	1.405	1.407					1.40	0.01	11.08	OK
				23	6	-0.1	6.00	5.90		5.932	5.952	5.948	5.951	5.948					5.95	0.01	2.05	OK
				24	R3.1	±0.5	3.60	2.60		3.085	3.064	3.053	3.056	3.079					3.07	0.01	11.84	OK
				25	R4.1	±0.5	4.60	3.60		3.911	3.892	3.942	3.940	3.949					3.93	0.02	6.86	OK
				26	R0.5	±0.3	0.80	0.20		4.120	4.036	4.089	4.055	4.024					4.06	0.04	2.54	OK
				27	R1	±0.3	1.30	0.70		1.114	1.074	1.023	1.051	1.013					1.06	0.04	2.45	OK
				28	R0.5	±0.3	0.80	0.20		0.522	0.545	0.546	0.534	0.533					0.54	0.01	10.13	OK
				29	R4.1	±0.5	4.60	3.60		3.810	3.791	3.801	3.826	3.792					3.80	0.01	11.49	OK
				30	R0.5	±0.3	0.80	0.20		0.683	0.697	0.679	0.699	0.693					0.69	0.01	11.38	OK
				31	R0.5	±0.3	0.80	0.20		0.337	0.344	0.341	0.357	0.330					0.34	0.01	10.02	OK
				32	R2	±0.5	2.50	1.50		1.854	1.877	1.868	1.870	1.892					1.87	0.01	12.02	OK
				33	Ø23	+0.2/-0.1	23.20	22.90		23.082	23.085	23.077	23.066	23.071					23.08	0.01	6.42	OK
				34	9.5	±1	10.50	8.50		9.716	9.884	9.654	9.518	9.903					9.74	0.16	2.06	OK
				35	Ø2	±0.2	2.20	1.80		2.092	2.093	2.079	2.095	2.081					2.09	0.01	8.99	OK
				36	R0.5	±0.3	0.80	0.20		0.254	0.250	0.275	0.264	0.292					0.27	0.02	5.88	OK
				37	R0.5	±0.3	0.80	0.20		0.263	0.233	0.267	0.247	0.247					0.25	0.01	7.28	OK
				38	3.5	+0.1/0	3.6	3.5		3.539	3.552	3.549	3.534	3.529					3.54	0.01	1.71	OK
* Control Code : Control Method Classification For the Inspection C:Special Characteristic, I:Important Inspection S:SPC(Statistical Process Control)				Pass rate =	Pass Point Total Point																	

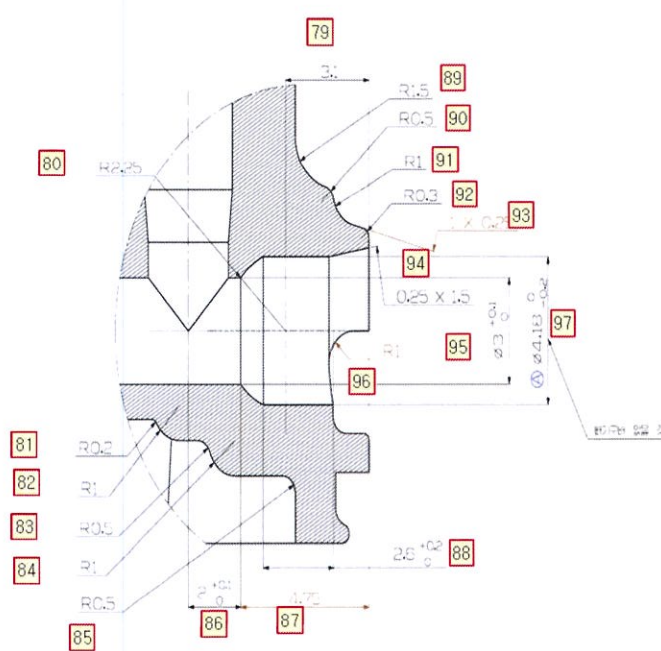
DIMENSION MEASUREMENT REPORT(B) Cavity #2

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance		Control Code	DATA SHEET													
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Compensation	USL		LSL	Date : 2021 / 07 / 02												
								X ₁	X ₂	X ₃	X ₄	X ₅					$\bar{X}$	R	Cp	판정	
Sketch for inspection	39	Ø8	±0.2	8.20	7.80		7.850	7.860	7.843	7.854	7.841					7.85	0.01	8.51	OK		
	40	R0.2	±0.3	0.50	0.00		0.219	0.208	0.208	0.221	0.227					0.22	0.01	9.94	OK		
	41	MIN2	-2	2.00	0.00		1.894	1.911	1.943	1.971	1.933					1.93	0.03	11.24	OK		
	42	16.7°	±1	17.70	15.70		17.252	17.413	17.214	16.924	17.256					17.21	0.18	1.87	OK		
	43	R0.2	±0.3	0.50	0.00		0.151	0.144	0.149	0.161	0.139					0.15	0.01	10.09	OK		
	44	7°	±1	8.00	6.00		7.122	7.249	7.104	7.304	7.306					7.22	0.10	3.41	OK		
	45	R0.2	±0.3	0.50	0.00		0.168	0.162	0.174	0.155	0.179					0.17	0.01	8.77	OK		
	46	3	±0.2	3.20	2.80		3.125	3.116	3.124	3.115	3.123					3.12	0.00	14.12	OK		
	47	Ø7.9	±0.1	8.00	7.80		7.851	7.842	7.865	7.858	7.854					7.85	0.01	3.91	OK		
	48	Ø6.8	±0.1	6.90	6.70		6.749	6.752	6.749	6.748	6.743					6.75	0.00	10.19	OK		
	49	Ø3.3	±0.1	3.40	3.20		3.323	3.310	3.312	3.315	3.326					3.32	0.01	4.78	OK		
	50	6	±0.2	6.20	5.80		6.180	6.164	6.169	6.193	6.157					6.17	0.01	4.71	OK		
	51	15	±0.25	15.25	14.75		15.002	15.010	15.001	14.992	15.024					15.01	0.01	6.94	OK		
	52	9	±0.2	9.20	8.80		8.832	8.848	8.828	8.855	8.809					8.83	0.02	3.70	OK		
	53	R0.1	-0.1	0.10	0.00		0.079	0.074	0.077	0.061	0.063					0.07	0.01	2.02	OK		
	54	4.5	±0.2	4.70	4.30		4.310	4.295	4.302	4.291	4.327					4.31	0.01	4.67	OK		
	55	Ø9.1	±0.2	9.3	8.9		9.140	9.132	9.130	9.161	9.138					9.14	0.01	5.40	OK		

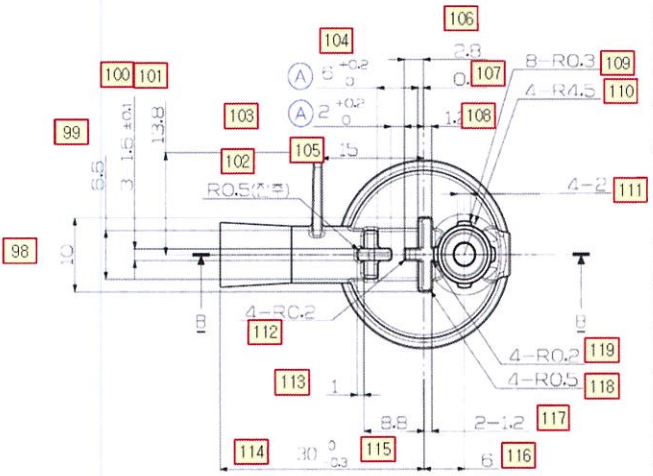
DIMENSION MEASUREMENT REPORT(B) Cavity #2

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance		Control Code	DATA SHEET														
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Master: / /	USL		LSL	Date : 2021 / 07 / 02	X ₁	X ₂	X ₃	X ₄	X ₅				$\bar{X}$	R	Cp	판정	
Sketch for inspection 				56	R0.2	±0.3	0.50	0.00		0.151	0.171	0.152	0.150	0.156				0.16	0.01	9.59	OK	
				57	1	±0.2	1.20	0.80		1.100	1.084	1.103	1.099	1.108				1.10	0.01	7.42	OK	
				58	1	±0.2	1.20	0.80		0.995	1.012	1.001	0.994	0.994				1.00	0.01	8.63	OK	
				59	R5	±0.5	5.50	4.50		4.916	4.949	4.958	4.967	4.968				4.95	0.02	7.81	OK	
				60	R6	±0.5	6.50	5.50		6.015	6.040	6.058	6.040	6.044				6.04	0.02	10.74	OK	
				61	Ø3	+0.1	3.10	3.00		3.051	3.069	3.061	3.062	3.046				3.06	0.01	1.81	OK	
				62	1.5x0.15	±0.2	1.70	1.30		1.302x0.137	1.401x0.141	1.394x0.145	1.355x0.138	1.371x0.144								OK
				63	4	±0.2	4.20	3.80		4.011	4.018	4.024	3.998	4.006				4.01	0.01	6.58	OK	
				64	6	±0.1	6.10	5.90		5.928	5.937	5.941	5.928	5.923				5.93	0.01	4.52	OK	
				65	Ø3	+0.1	3.10	3.00		3.055	3.048	3.063	3.043	3.059				3.05	0.01	2.05	OK	
				66	4.5	±0.1	4.60	4.40		4.541	4.529	4.534	4.549	4.533				4.54	0.01	4.23	OK	
				67	0.7	-0.1	0.70	0.60		0.670	0.648	0.662	0.659	0.658				0.66	0.01	2.10	OK	
				68	6	-0.2	6.00	5.80		5.824	5.825	5.847	5.801	5.825				5.82	0.02	2.05	OK	
				69	Ø20	+0.2	20.20	20.00		20.066	20.085	20.073	20.061	20.072				20.07	0.01	3.70	OK	
				70	Ø23	+0.2/-0.1	23.20	22.90		23.095	23.084	23.081	23.073	23.094				23.09	0.01	5.41	OK	
				71	Ø24	+0.2/-0.1	24.20	23.90		23.932	23.945	23.924	23.943	23.936				23.94	0.01	5.87	OK	
				72	Ø25.5	-0.1	25.50	25.40		25.431	25.438	25.442	25.436	25.444				25.44	0.01	3.26	OK	
73	1.1	±0.2	1.30	0.90		1.180	1.160	1.173	1.157	1.161				1.17	0.01	6.78	OK					
74	2	-0.1	2.00	1.90		1.949	1.952	1.949	1.967	1.956				1.95	0.01	2.22	OK					
75	1	±0.2	1.20	0.80		1.143	1.154	1.120	1.160	1.120				1.14	0.02	3.56	OK					
76	R0.5	±0.3	0.80	0.20		0.517	0.518	0.530	0.510	0.500				0.52	0.01	9.05	OK					
77	R0.25	±0.3	0.55	0.00		0.262	0.255	0.260	0.243	0.271				0.26	0.01	8.92	OK					
78	Rz3.2		3.20	0.00		0.771	0.694	0.518	0.711	0.656				0.67	0.09	5.64	OK					
* Control Code : Control Method Classification For the Inspection C:Special Characteristic, I:Important Inspection S:SPC(Statistical Process Control)				Pass rate =		Pass Point																
						Total Point																

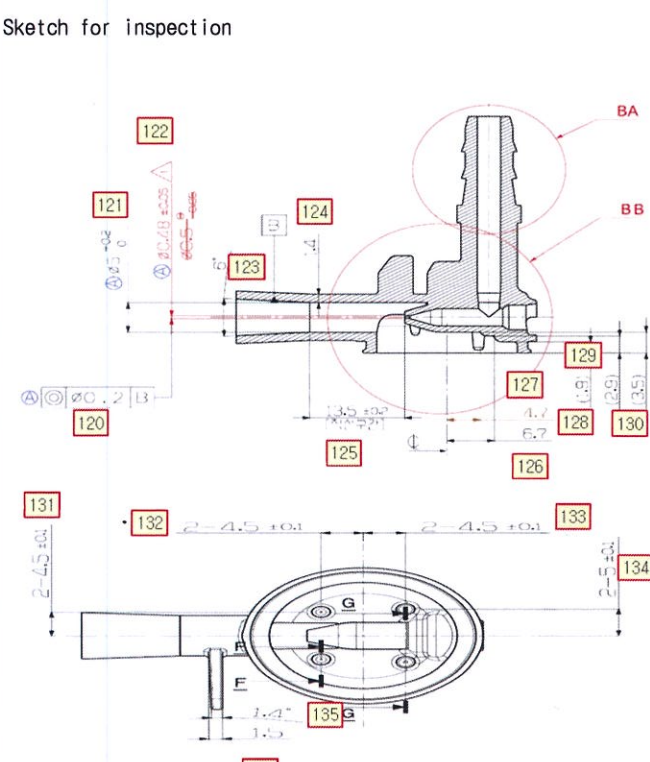
DIMENSION MEASUREMENT REPORT(B) Cavity #2

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance			Control Code	DATA SHEET																	
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Compensation	USL	LSL		Date : 2021 / 07 / 02																	
														X ₁	X ₂	X ₃	X ₄	X ₅					$\bar{X}$	R	Cp	판정
Sketch for inspection 				79	3.1	±0.2	3.30	2.90		3.183	3.188	3.194	3.188	3.174					3.19	0.01	8.92	OK				
				80	R2.25	±0.5	2.75	1.75		2.005	2.012	2.031	2.010	1.994					2.01	0.01	12.38	OK				
				81	R0.2	±0.3	0.50	0.00		0.284	0.302	0.294	0.311	0.307					0.30	0.01	7.73	OK				
				82	R1	±0.3	1.30	0.70		1.007	0.994	1.000	0.984	1.013					1.00	0.01	8.86	OK				
				83	R0.5	±0.3	0.80	0.20		0.534	0.533	0.544	0.521	0.525					0.53	0.01	11.23	OK				
				84	R1	±0.3	1.30	0.70		1.148	1.154	1.139	1.133	1.140					1.14	0.01	12.15	OK				
				85	R0.5	±0.3	0.80	0.20		0.421	0.411	0.430	0.418	0.424					0.42	0.01	14.18	OK				
				86	2	+0.1	2.10	2.00		2.083	2.075	2.076	2.094	2.086					2.08	0.01	2.14	OK				
				87	4.75	±0.2	4.95	4.55		4.699	4.698	4.709	4.693	4.694					4.70	0.01	10.50	OK				
				88	2.6	+0.2	2.80	2.60		2.632	2.626	2.628	2.632	2.641					2.63	0.01	5.79	OK				
				89	R1.5	±0.3	1.80	1.20		1.433	1.444	1.439	1.434	1.436					1.44	0.00	22.53	OK				
				90	R0.5	±0.3	0.80	0.20		0.498	0.500	0.481	0.498	0.510					0.50	0.01	9.59	OK				
				91	R1	±0.3	1.30	0.70		0.854	0.841	0.847	0.868	0.867					0.86	0.01	8.35	OK				
				92	R0.3	±0.3	0.60	0.00		0.367	0.356	0.370	0.382	0.365					0.37	0.01	10.63	OK				
				93	1X0.25	±0.2				0.984x0.244	0.971x0.249	0.966x0.261	0.949x0.255	0.952x0.231								OK				
				94	0.25X1.5	±0.2				0.236x1.487	0.224x1.499	0.241x1.501	0.236x1.493	0.235x1.439								OK				
				95	Ø3	+0.1	3.10	3.00		3.063	3.061	3.050	3.045	3.056					3.06	0.01	2.22	OK				
				96	R1	±0.3	1.30	0.70		0.854	0.834	0.874	0.867	0.857					0.86	0.02	6.57	OK				
				97	Ø4.18	-0.2	4.18	3.98		4.129	4.129	4.119	4.126	4.121					4.12	0.00	7.24	OK				
* Control Code : Control Method Classification For the Inspection C:Special Characteristic, I:Important Inspection S:SPC(Statistical Process Control)				Pass rate =	Pass Point	Total Point																				

DIMENSION MEASUREMENT REPORT(B) Cavity #2

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance		Control Code	DATA SHEET																
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Compensation	Master : / /		Date : 2021 / 07 / 02																
						USL		LSL	X ₁	X ₂	X ₃	X ₄	X ₅					$\bar{X}$	R	Cp	판정			
Sketch for inspection 				98	10	±0.25	10.25	9.75		10.092	10.075	10.094	10.093	10.086				10.09	0.01	10.54	OK			
				99	6.6	±0.25	6.80	6.40		6.634	6.615	6.616	6.641	6.643				6.63	0.01	4.95	OK			
				100	1.6	±0.1	1.70	1.50		1.677	1.629	1.654	1.671	1.649				1.66	0.02	1.75	OK			
				101	13.8	±0.25	14.05	13.55		13.835	13.811	13.840	13.869	13.842				13.84	0.02	4.03	OK			
				102	R0.5	±0.3	0.80	0.20		0.484	0.478	0.475	0.461	0.483				0.48	0.01	10.80	OK			
				103	2	+0.2	2.20	2.00		2.145	2.144	2.153	2.141	2.122				2.14	0.01	2.90	OK			
				104	6	+0.2	6.20	6.00		6.133	6.126	6.131	6.124	6.135				6.13	0.00	7.16	OK			
				105	15	±0.25	15.25	14.75		15.040	15.051	15.042	15.037	15.049				15.04	0.01	13.95	OK			
				106	2.8	±0.2	3.00	2.60		2.605	2.589	2.608	2.606	2.599				2.60	0.01	8.66	OK			
				107	0.8	±0.2	1.00	0.60		0.663	0.663	0.665	0.663	0.652				0.66	0.01	12.78	OK			
				108	1.2	±0.2	1.40	1.00		1.329	1.319	1.334	1.330	1.337				1.33	0.01	9.76	OK			
				109	R0.3	±0.3	0.60	0.00		0.326	0.323	0.326	0.308	0.309				0.32	0.01	10.96	OK			
				110	R4.5	±0.5	5.0	4.0		4.553	4.463	4.558	4.550	4.565				4.54	0.04	3.95	OK			
				111	2	±0.2	2.20	1.80		2.112	2.111	2.101	2.102	2.093				2.10	0.01	8.49	OK			
				112	R0.2	±0.3	0.50	0.00		0.247	0.240	0.220	0.220	0.260				0.24	0.02	4.78	OK			
				113	1	±0.2	1.2	0.8		0.987	0.993	1.000	0.985	0.981				0.99	0.01	8.97	OK			
				114	30	-0.3	30.0	29.7		29.725	29.715	29.746	29.728	29.704				29.72	0.02	3.19	OK			
				115	8.8	±0.2	9.0	8.6		8.798	8.799	8.781	8.820	8.785				8.80	0.02	4.36	OK			
				116	6	±0.2	6.2	5.8		6.098	6.081	6.087	6.087	6.106				6.09	0.01	6.64	OK			
				117	1.2	±0.2	1.4	1.0		1.301	1.324	1.281	1.289	1.282				1.30	0.02	3.73	OK			
118	R0.5	±0.3	0.8	0.2		0.419	0.441	0.434	0.398	0.416				0.42	0.02	5.96	OK							
119	R0.2	±0.3	0.5	0.0		0.286	0.269	0.293	0.303	0.281				0.29	0.01	6.53	OK							
※ Control Code : Control Method Classification For the Inspection C:Special Characteristic, I:Important Inspection S:SPC(Statistical Process Control)				Pass rate = $\frac{\text{Pass Point}}{\text{Total Point}}$																				

DIMENSION MEASUREMENT REPORT(B) Cavity #2

Part No.	1058-N90260-10	Inspection Area		SPEC	Tolerance		Control Code	DATA SHEET													
Part Name	ASSIST-PUMP ASSY-SUB	Usage	No		Master : / /	USL		LSL	Date : 2021 / 07 / 02												
				Compensation				X ₁	X ₂	X ₃	X ₄	X ₅					$\bar{X}$	R	Cp	판정	
Sketch for inspection 				120	◎0.2		0.20	0.00		0.120	0.109	0.130	0.123	0.114				0.12	0.01	4.11	OK
				121	Ø5	+0.2	0.20	0.00		5.101	5.104	5.101	5.090	5.092				5.10	0.01	5.39	OK
				122	Ø0.48	±0.05	0.53	0.43		0.473	0.473	0.475	0.480	0.472				0.47	0.00	5.19	OK
				123	6°	±1	7.00	5.00		5.964	5.844	5.912	5.936	5.859				5.90	0.05	6.57	OK
				124	1.4	±0.2	1.60	1.20		1.429	1.421	1.442	1.436	1.432				1.43	0.01	8.50	OK
				125	13.5	±0.2	13.70	13.30		13.465	13.453	13.470	13.448	13.482				13.46	0.01	4.91	OK
				126	6.7	±0.2	6.90	3.50		6.539	6.584	6.631	6.652	6.636				6.61	0.05	12.23	OK
				127	4.7	±0.2	4.90	3.50		4.636	4.649	4.681	4.722	4.739				4.69	0.04	5.22	OK
				128	1.9	±0.2	2.10	1.70		1.949	1.941	1.938	1.951	1.958				1.95	0.01	8.31	OK
				129	2.9	±0.2	3.10	2.70		2.882	2.887	2.872	2.883	2.885				2.88	0.01	11.48	OK
				130	3.5	±0.2	3.70	3.30		3.671	3.683	3.667	3.677	3.673				3.67	0.01	10.93	OK
				131	4.5	±0.1	4.60	4.40		4.595	4.614	4.610	4.582	4.600				4.60	0.01	2.63	OK
				132	4.5	±0.1	4.60	4.40		4.484	4.471	4.491	4.469	4.493				4.48	0.01	3.00	OK
				133	4.5	±0.1	4.60	4.40		4.560	4.539	4.547	4.546	4.563				4.55	0.01	3.29	OK
				134	5	±0.1	5.10	4.90		5.073	5.060	5.072	5.093	5.083				5.08	0.01	2.68	OK
				135	1.4°	±1	2.40	0.40		1.313	1.211	1.104	1.229	1.239				1.22	0.08	4.43	OK
136	1.5	±0.2	1.70	1.30		1.435	1.436	1.446	1.416	1.439				1.43	0.01	5.98	OK				
															</						

● 재질 시험 결과 성적서

※ 첨부 위치 : 재질성적서 서류 시작점 (QMS)

차종 : AX

품명 : ASSIST PUMP ASSY -SUB

품번 : 1058-N90260-10

업체명 : 승정

결 재	작성	검토	승인
			

[illegible]

BODY(POM F25-03)

재질성적서

(오방)

문서번호: Q210117
울산공장 / 품질팀 (052-279-7831)
오형석 (hyeongseok.oh@kepital.com)

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Tel. 052-279-7830~4 Fax. 052-279-7889
품질팀장 박장호 (jangho.park@kepital.com)

수신 : 현답산업

2021.01.17

참조 : 오방

제목 : KEPITAL F25-03 NAT 물성 평가 결과

귀사의 일익 번창하심을 기원 하오며, 귀사에서 요청하신 KEPITAL F25-03 NAT 의 물성 평가 결과를 다음과 같이 송부 드리오니, 업무에 참조하시기 바랍니다.

- 다 음 -



1. 요청 사항 : KEPITAL F25-03 NAT 물성 평가

2. 평가 결과

구분	시험방법	단위	MS237-09 Type A	측정치	비고
용융지수	ISO 1133	g/10min	11.0 ~ 15.0	13.0	
인장강도	ISO 527	MPa	Min 57	64.9	속도 : 50 mm/min
신율	ISO 527	%	Min 40	55	속도 : 50 mm/min
굴곡강도	ISO 178	MPa	Min 74	88.0	속도 : 2 mm/min
굴곡탄성률	ISO 178	MPa	Min 2150	2,540	속도 : 2 mm/min
IZOD 충격강도 (23°C)	ISO 180/A	kJ/m²	Min 6.6	6.6	
IZOD 충격강도 (-30°C)	ISO 180/A	kJ/m²	Min 5.8	5.9	
록크웰경도	ISO 2039-2	M Scale	Min 80	86	
열변형온도 (고하중)	ISO 75	°C	Min 78	100	하중 : 1.8 MPa
연소성 (HMC)	MS300-08	mm/min	Max 80	32	
비중	ISO 1183	-	1.39 ~ 1.43	1.41	
융점	ISO 11357	°C	Min 162	166	
열중량분석	ISO 11358	%	-	0.15	2/4 페이지 참조
시차주사열량분석	ISO 11357	°C	-	165.9	3/4 페이지 참조
적외선 분광분석	FT-IR (ATR)	-	-	-	4/4 페이지 참조

※ KEPITAL F25-03 은 HMC 미등록 Grade 이며, 고객 요청사항에 의거 MS237-09 시험법을 바탕으로 평가된 1회성 자료입니다.

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<주의> 상기 자료는 본 제품에 대해 당사의 현재 기술 수준에서 측정된 것이며, 측정 방법 및 조건에 따라 변경될 수 있습니다. 본 제품에 고객에 의해 안료 및 기타 첨가제가 사용된 경우 상기 자료는 적용되지 않습니다. 본 제품은 (치)의학 Implants 용에는 적합하지 않으며, 고객은 안전 및 보건 기준에 따라 본 제품을 사용해야 합니다. 제품 사용의 결정 및 책임은 고객에게 있으며, 상기 자료는 법적 소송 및 근거자료로 활용될 수 없습니다.

A. 열중량 분석 (TGA)

(1) 시험 조건

- MS237-09 규격에 따라 평가 실시함.

※ 세부 조건

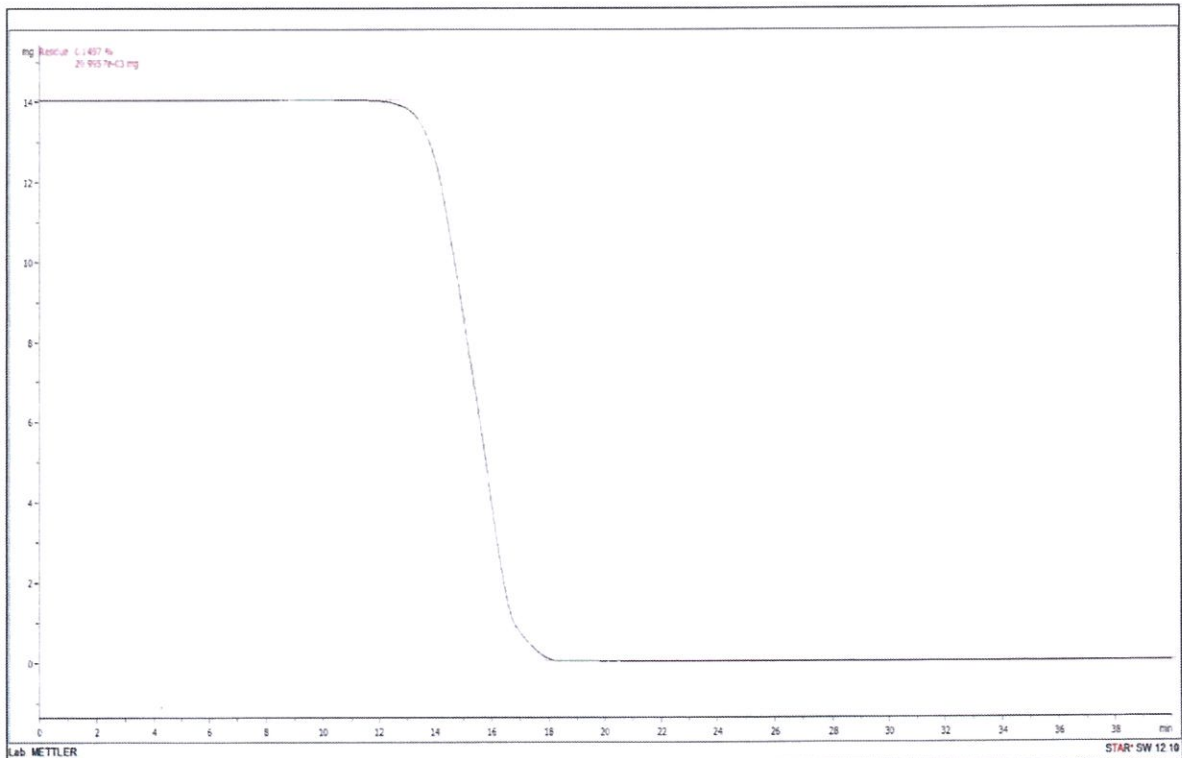
- Temperature : 40 ~ 600 °C (N₂), 600~850 °C (Air)

- Rate : 20°C/min

(2) 평가 시료 : KEPITAL F25-03 NAT

(3) 평가 결과

- Ash 잔량 : 0.15%



B. 시차주사열량분석 (DSC)

(1) 시험 조건

- MS237-09 규격에 따라 평가 실시함.

※ 세부 조건

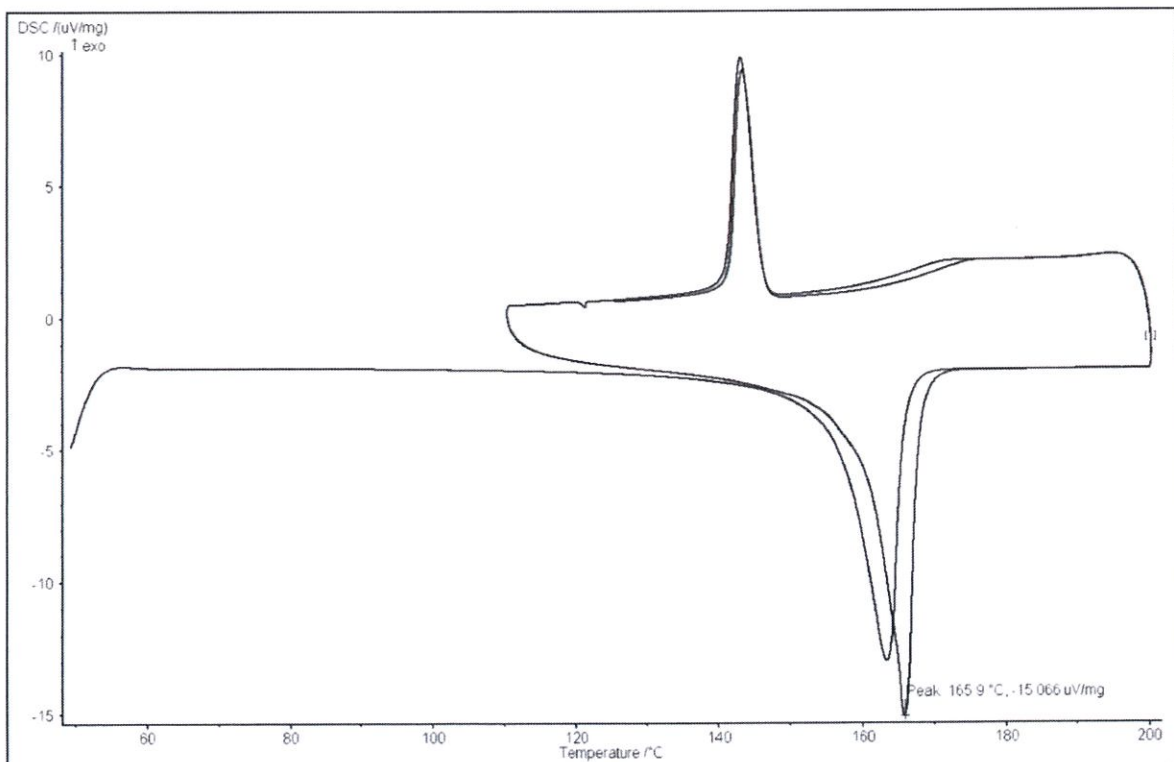
① 50 ~ 200 °C, 20 °C/분 (승온)

② 200 ~ 50 °C, 20 °C/분 (냉각)

(2) 평가 시료 : KEPITAL F25-03 NAT

(3) 평가 결과

- Tm (°C) : 165.9°C



C. 적외선 분광분석 (FT-IR)

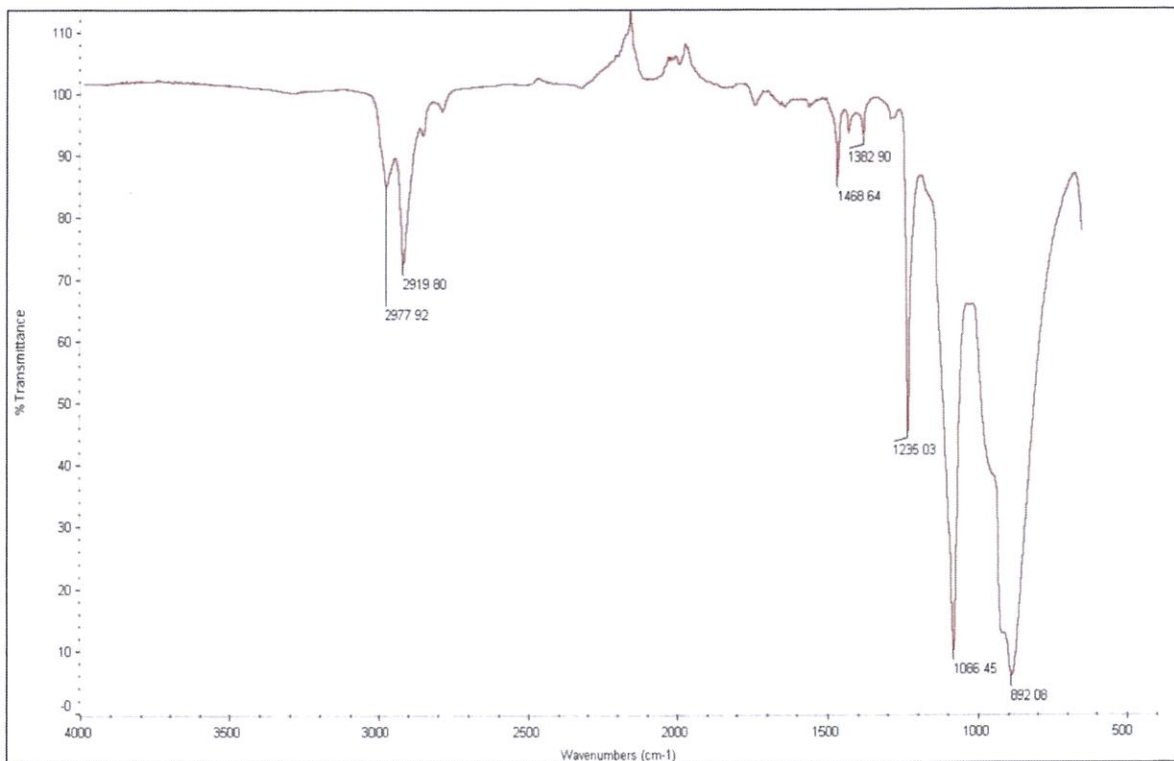
(1) 평가 조건

- MS237-09 규격에 따라 평가 실시함.

※ 세부 조건 : 측정범위 4000 ~ 650 cm^{-1} , 스캔수 32 회, 분해능 4 cm^{-1}

(2) 평가 시료 : KEPITAL F25-03 NAT

(3) 평가 결과



[Handwritten Signature]

한국엔지니어링플라스틱(주)

품질팀장 박장호

O-RING(FKM)

재질성적서

(삼주기업)



DAIKIN KOREA CO., LTD.

WOO SHIN CHEMTECH CORP

UNEDA CENTER BLDG., 4-12 MAKAZAKI-NISHI
2-CHOME KITA-KU, OSAKA 530-8323 JAPAN
TEL. 81-6-6374-9300 FAX. 81-6-6373-4281

QUALITY CONTROL REPORT

We hereby state that the following material is in accordance with our specifications.

Date : February, 5. 2021
Product : DAI-EL C-621
Lot No. : DL6BV11001

Contents : 20 kg x 25
Ref No. : P02020120008-2-1

Inspection Item	Unit	Data	Test method * Specification
Appearance		Good	Visual inspection White to light pink
Mooney viscosity		77	JIS K 6300 (MLI+10 100°C) 80 to 93
Cure time (t90)	min	5.7	JSR CURELASTO-METER 170°C 2.5 to 7.5
Tensile strength Type 5	MPa	15.3	JIS K 6251 12.0 to 20.0
Elongation Type 5	%	330	JIS K 6251 210 to 380
Hardness (SHORE A)		73	JIS K 6253 (Peak) 71 to 81



Note

DAIKIN INDUSTRIES, LTD.

M. Karamitsu

Section Manager
Quality Assurance Department
Chemicals Division
MASAKI KURAHITSU

※ 첨부 위치 : 중금속 서류 시작점 (QMS)

결 재	작성	검토	승인
	김영 영		

[illegible]

BODY(POM F25-03)

중금속성적서

(오방)



업체 : (주)오방
 담당 : 수지사업부 임종섭 이사
 재질 : POM
 Grade : F10-03H NAT, F20-03 NAT, F25-03 NAT, F10-52H NAT
 판정 : 합격

원본대조필



Test Report No. F690101/LF-CTSAYGA21-00152

Issued Date : 2021. 01. 13

Page 1 of 8

KOREA ENGINEERING PLASTICS CO., LTD.

84, Jangsaengpogorae-ro
 Nam-gu, Ulsan
 Korea



The following sample(s) was/were submitted and identified by/on behalf of the client as:-

SGS File No. : AYGA21-00152
Product Name : KEPITAL F10-02
Item No./Part No. : KEPITAL F10-02
Client Reference Data : F10-01, F10-02, F10-03H(M/V), F10-03H LOF(2), F10-01
 T2, F10-52(H/D), F10-52HLOF(2), F15-33(HC), F15-33LOF(2), F20-01, F20-02, F20-03, F20-03M, F2
 0-03LOF(2), F20-33, F20-52(G), F20-52LOF(2), F20-52G LOF(2), F25-03(M), F25-03
 LOF(2), F25-03H(M), F25-03H LOF(2), F25-03HT, F25-03HT
 LOF(2), F25-63, F30-01(M), F30-03(M), F30-03 LOF(2) F30-34
 (LOF(2)), F30-52(G/LOF(2)), F40-03(M/LOF(2)), F40-34 (LOF), TX-11H, TX-11H
 LOF(2), TX-11HK, TX-11HK
 LOF(2), TX-21(LOF(2)), TX-31(LOF(2)), MX20BT01, MX25BT01, MX30BT01, MX25BT03, MX25LF0
 1, TX-81DT, AR-10(S), AR-15, AR-20, F20-03HC, F20-03HC LOF(2), F30-03HC(LOF), F20-03M
 LOF, F25-03M LOF, F30-03M LOF, F30-34M LOF
Received Date : 2021. 01. 06
Test Period : 2021. 01. 06 to 2021. 01. 13
Report Comments : By the applicant's request, item No.s/part No.s & client reference information are stated/added on
 report.
Test Results : For further details, please refer to following page(s)

SGS Korea Co., Ltd.

Tommy

Tommy Oh / Chemical Lab Mgr

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CQP-7000F07 (01)



SGS Korea Co., Ltd.

100, The Gateway 7F, LS10, Dongsongu, Anyangsi, Gyeonggi-do, Korea 141-710
 T+82 (0)31 4506 000 F+82 (0)31 4506 055 <http://www.sgs.co.kr>

Membre du réseau SGS Group de Certification de Conformité



업체 : (주)오방
 담당 : 수지사업부 임종섭 이사
 재질 : POM
 Grade : F10-03H NAT, F20-03 NAT, F25-03 NAT, F10-52H NAT
 판정 : 합격

원본대조필



Test Report No. F690101/LF-CTSAYGA21-00152

Issued Date : 2021. 01. 13

Page 2 of 8

Sample No. : AYGA21-00152.001
 Sample Description : KEPITAL F10-02
 Item No./Part No. : KEPITAL F10-02
 Materials : POM

Heavy Metals

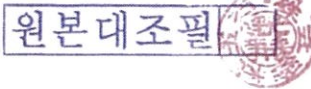
Test Items	Unit	Test Method	MDL	Results
Cadmium (Cd)	mg/kg	With reference to IEC 62321-5 : 2013, by ICP-OES	0.5	N.D.
Lead (Pb)	mg/kg	With reference to IEC 62321-5 : 2013, by ICP-OES	5	N.D.
Mercury (Hg)	mg/kg	With reference to IEC 62321-4 : 2013+A1 : 2017, by ICP-OES	2	N.D.
Hexavalent Chromium (Cr VI)*	mg/kg	With reference to IEC 62321-7-2 : 2017, by UV-Vis and/or with reference to IEC 62321-5 : 2013, by ICP-OES	8	N.D.
Antimony (Sb)	mg/kg	With reference to EPA 3052 : 1996, EPA 6010D : 2018, by ICP	10	N.D.

Flame Retardants-PBBs/PBDEs

Test Items	Unit	Test Method	MDL	Results
Monobromobiphenyl	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Dibromobiphenyl	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Tribromobiphenyl	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Tetrabromobiphenyl	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Pentabromobiphenyl	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Hexabromobiphenyl	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Heptabromobiphenyl	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Octabromobiphenyl	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Nonabromobiphenyl	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Decabromobiphenyl	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Monobromodiphenyl ether	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Dibromodiphenyl ether	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Tribromodiphenyl ether	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Tetrabromodiphenyl ether	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Pentabromodiphenyl ether	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Hexabromodiphenyl ether	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Heptabromodiphenyl ether	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Octabromodiphenyl ether	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Nonabromodiphenyl ether	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.
Decabromodiphenyl ether	mg/kg	With reference to IEC 62321-6 : 2015, by GC-MS	5	N.D.

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업체 : (주)오방
 담당 : 수직사업부 임종섭 이사
 제품 : POM
 Grade : F10-03H NAT, F20-03 NAT, F25-03 NAT, F10-52H NAT
 판정 : 합격



Test Report No. F690101/LF-CTSAYGA21-00152

Issued Date : 2021. 01. 13

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Sample No. : AYGA21-00152.001
 Sample Description : KEPITAL F10-02
 Item No./Part No. : KEPITAL F10-02
 Materials : POM

Phthalates

Test Items	Unit	Test Method	MDL	Results
Di-(2-ethylhexyl) phthalate (DEHP)	mg/kg	With reference to IEC 62321-8 : 2017, by GC-MS	50	N.D.
Di-butyl phthalate (DBP)	mg/kg	With reference to IEC 62321-8 : 2017, by GC-MS	50	N.D.
Benzyl butyl phthalate (BBP)	mg/kg	With reference to IEC 62321-8 : 2017, by GC-MS	50	N.D.
Di-isobutyl phthalate (DIBP)	mg/kg	With reference to IEC 62321-8 : 2017, by GC-MS	50	N.D.
Di-isononyl phthalate (DINP)	mg/kg	With reference to IEC 62321-8 : 2017, by GC-MS	50	N.D.
Di-isodecyl phthalate (DIDP)	mg/kg	With reference to IEC 62321-8 : 2017, by GC-MS	50	N.D.
Di-n-hexyl phthalate (DNHP)	mg/kg	With reference to IEC 62321-8 : 2017, by GC-MS	50	N.D.
Di-n-octyl phthalate (DNOP)	mg/kg	With reference to IEC 62321-8 : 2017, by GC-MS	50	N.D.

Chlorinated Paraffin

Test Items	Unit	Test Method	MDL	Results
Alkanes, C14~17, Medium Chain Chlorinated Paraffins(MCCP)	mg/kg	With reference to ISO 18219, by CI-GC-MS	50	N.D.

Halogen Content

Test Items	Unit	Test Method	MDL	Results
Chlorine(Cl)	mg/kg	With reference to EN 14582 : 2016, by IC	30	N.D.

Flame Retardants

Test Items	Unit	Test Method	MDL	Results
Hexabromocyclododecane (HBCDD)	mg/kg	With reference to USEPA 3540 C, by LC/MS	5	N.D.

- NOTE: (1) N.D. = Not detected. (<MDL)
 (2) mg/kg = ppm, µg/kg = ppb, mg/L=ppm
 (3) MDL = Method Detection Limit
 (4) - = No regulation
 (5) Negative = Undetectable / Positive = Detectable
 (6) ** = Qualitative analysis (No Unit)
 (7) * = a. The result of Hexavalent Chromium (Cr(VI)) is "ND" as the result of Chromium (Cr) is "ND", and confirmation test of Hexavalent Chromium (Cr(VI)) is not required.
 b. If the Chromium (Cr) content is greater than the MDL of Hexavalent Chromium (Cr(VI)), confirmation test of Hexavalent Chromium (Cr(VI)) is required.
 (8) The results shown in this test report refer only to the sample(s) tested unless otherwise stated.
 This test report is not related to Korea Laboratory Accreditation Scheme.

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업체 : (주)오방
담당 : 수지사업부 임종섭 이사
제질 : POM
Grade : F10-03H NAT, F20-03 NAT, F25-03 NAT, F10-52H NAT
판정 : 합격

SGS

원본대조필

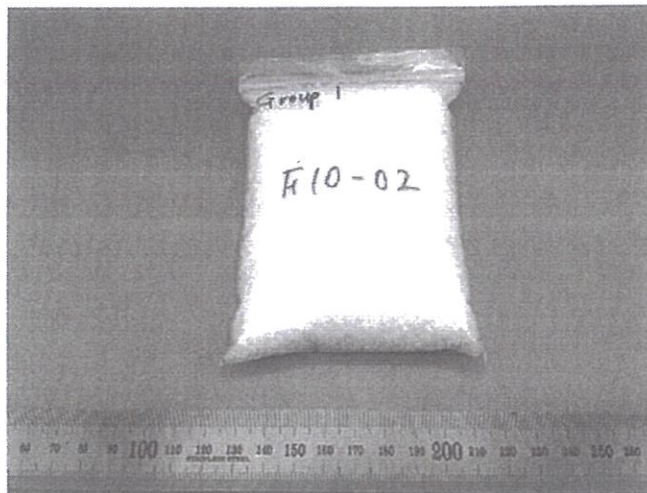


Test Report No. F690101/LF-CTSAYGA21-00152

Issued Date : 2021. 01. 13

Page 4 of 8

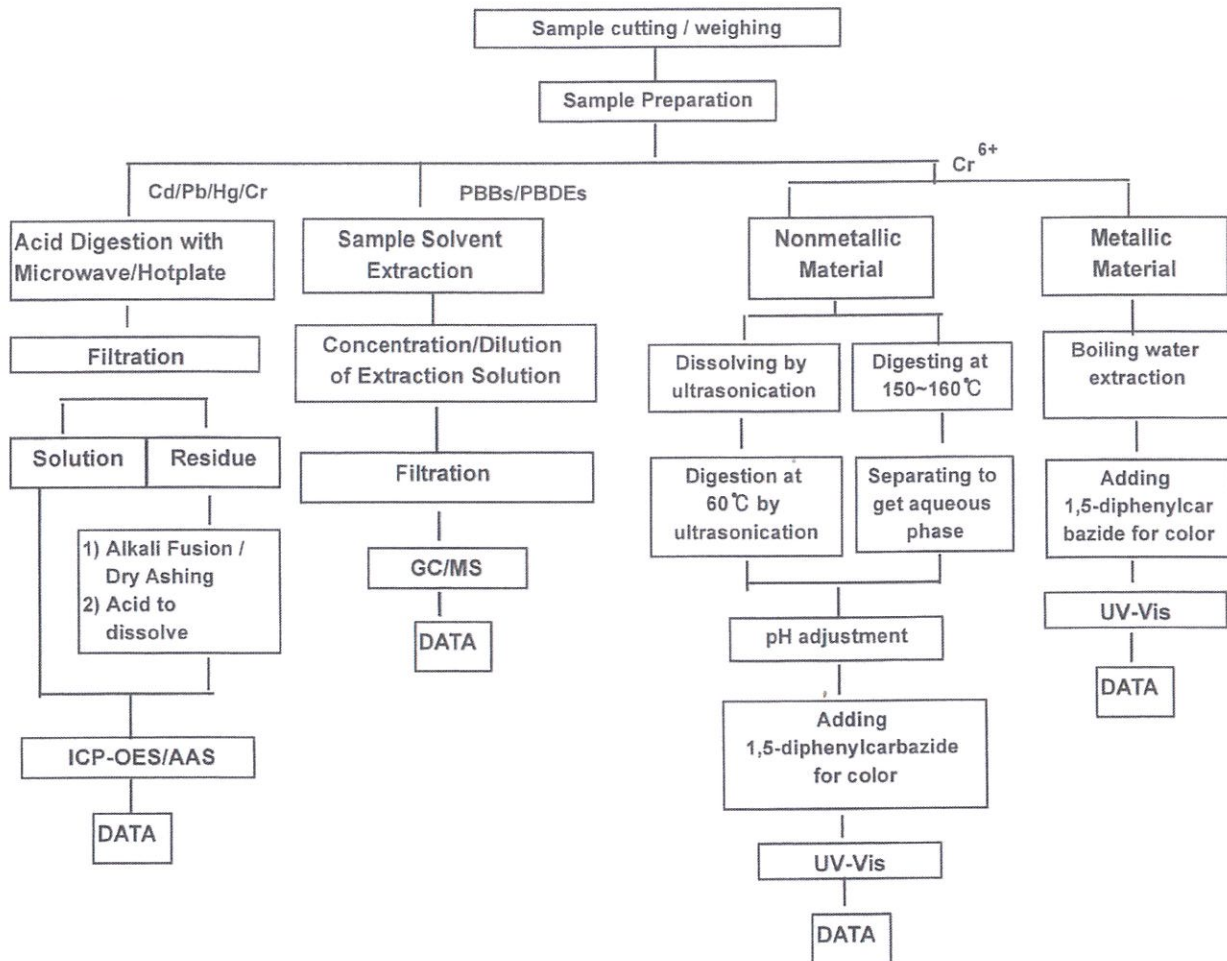
Picture of Sample as Received:



AYGA21-00152.001

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Testing Flow Chart for RoHS: Cd/Pb/Hg/Cr⁶⁺ /PBBs&PBDEs Testing



The samples were dissolved totally at the acid digestion step of the above flow chart for Cd,Pb,Hg
Section Chief : Timothy Jeon

업체 : (주)오방
담당 : 수지사업부 임종섭 이사
재질 : POM
Grade : F10-03H NAT, F20-03 NAT, F25-03 NAT, F10-52H NAT
판정 : 합격

SGS

원본대조필

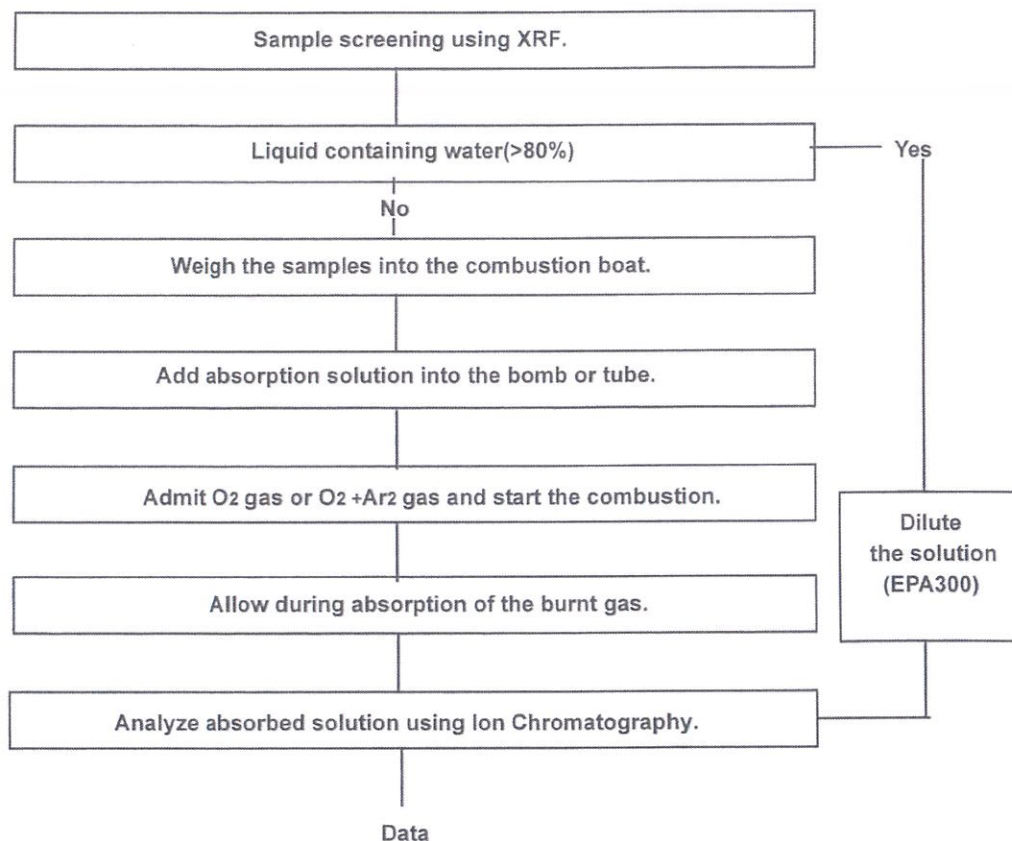


Test Report No. F690101/LF-CTSAYGA21-00152

Issued Date : 2021. 01. 13

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Flow Chart for Halogen Test



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업체 : (주)오빛
담당 : 수직사업부 임종섭 이사
재질 : POM
Grade : F10-03H NAT, F20-03 NAT, F25-03 NAT, F10-52H NAT
판정 : 합격

원본대조필

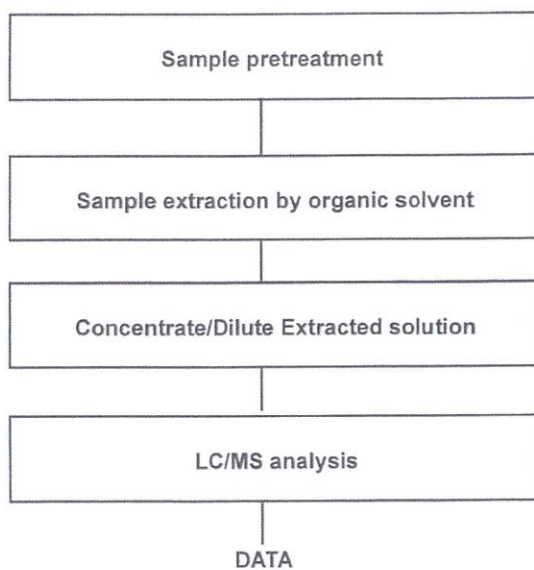


Test Report No. F690101/LF-CTSAYGA21-00152

Issued Date : 2021. 01. 13

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Testing Flow Chart for HBCD



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CQP-7081-F07 (01)

SGS Korea Co., Ltd.

332 Tte Gwangju, Jeonnam, 514-700, Republic of Korea
T: +82-61-4604-000 F: +82-61-4604-755 <http://www.sgs.co.kr>

Member of the SGS Group & part of the SGS Group

업체 : (주)오방
담당 : 수지사업부 임종섭 이사
재질 : POM
Grade : F10-03H NAT, F20-03 NAT, F25-03 NAT, F10-52H NAT
판정 : 합격

SGS

원본대조필

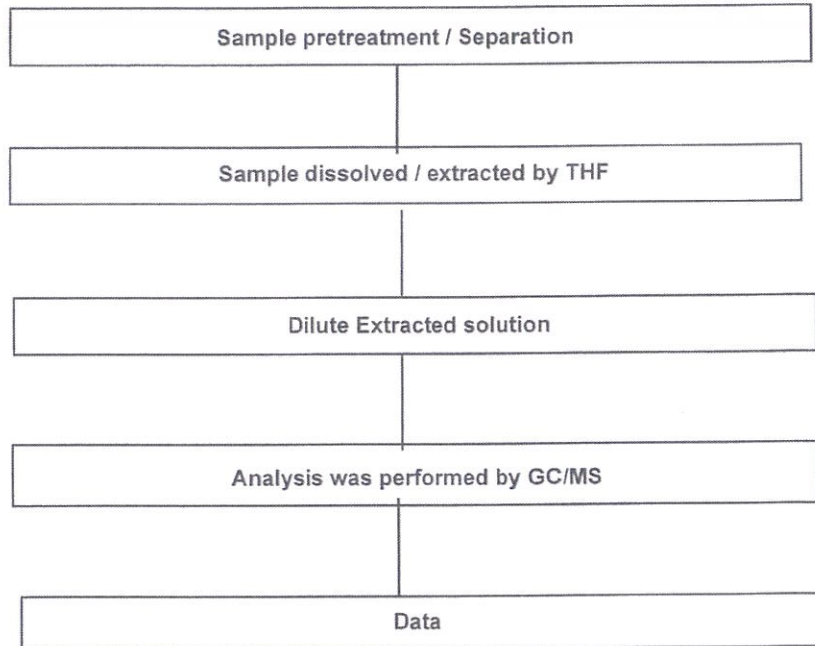


Test Report No. F690101/LF-CTSAYGA21-00152

Issued Date : 2021. 01. 13

Page 8 of 8

Flow Chart for Phthalate Test



*** End of Report ***

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CQP-7081-F07 (01)

SGS 한국지사

300 The Gateway To US is Our Gateway Anytime Anywhere Korea 14117
1-82-0201-6605-000 1-82-0201-4507-155 <http://www.sgs.co.kr>

Member of the SGS Group of Companies

O-RING(FKM)

중금속성적서

(삼주기업)

TEST REPORT
(시험 성적서)

신청기관 (인) : (주)삼주기업
APPLICANT : SAM JU INDUSTRY CO., LTD.

주소 (한글) : 대구광역시 달서구 대전동 618
ADDRESS (ENGL.) : 618, Daecheon-dong, Dalseo-gu,
Daegu, Korea

발행면수 (PAGE) : 1 of 5

발행일자 (DATE) : 2021. 02. 02.

시험성적서 번호 (REPORT NO.) : RT21R-S0694-004-K

시료 명세 : 시료에 대한 상세한 정보는 아래와 같음
(SAMPLE DESCRIPTION) (The following submitted sample(s) said to be)

제품명/형식 : FKM
(NAME/TYPE OF PRODUCT) (FKM)

시료고유번호 : RT21R-S0694-004
(SAMPLE ID NO.) (RT21R-S0694-004)

제품 생산자/공급자 : (주)삼주기업
(MANUFACTURER/VENDOR) (SAM JU INDUSTRY CO., LTD.)

시료접수일자 : 2021. 01. 28.
(SAMPLE RECEIVED) (Jan. 28, 2021)

시험일자 : 2021. 01. 28. ~ 2021. 02. 02.
(TESTING DATE) (Jan. 28, 2021 ~ Feb. 02, 2021)

시험방법 : 이 시험성적서의 다음 페이지 첨부
(TEST METHOD) (Please see the following page)

시험결과 : 이 시험성적서의 다음 페이지 첨부
(TEST RESULT) (Please see the following page)

비고 (Notes) : 1 이 시험성적서는 제시된 시료 및 시료명으로 시험한 결과로서 유사 대상시료에 적용할 수 없음
(The test results presented in this report refer only to the object tested)
2 이 시험성적서는 승인없이 복사 사용을 금함.
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승인자 (Approved by)

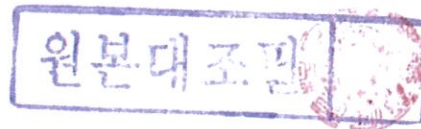
이응용 / 기술책임자
(E.Y. Lee / Lab. Technical Manager)

권한자 (Authorized by)

유환우 / 소장
(H.W. Yoo / Lab. General Manager)



Authenticity check

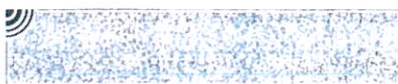


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TEST REPORT

(시험 성적서)

발행면수 (PAGE) : 2 of 5

발행일자 (DATE) : 2021. 02. 02.

시험성적서 번호 (REPORT NO.) : RT21R-S0694-004-K

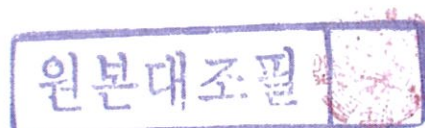
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시료명 (SAMPLE DESCRIPTION) : FKM
(FKM)

시험항목 (TEST ITEM)	단위 (UNIT)	분석방법 (TEST METHOD)	검출한계 (MDL)	시험결과 (RESULT)
카드뮴 (Cadmium, Cd)	mg/kg	With reference to IEC 62321-5 Edition 1.0 : 2013, by acid digestion and determined by ICP-OES	0.5	N.D.
납 (Lead, Pb)	mg/kg		5	N.D.
수은 (Mercury, Hg)	mg/kg	With reference to IEC 62321-4 : 2013/AMD1 : 2017, by acid digestion and determined by ICP-OES	2	N.D.
6가 크롬 (Hexavalent Chromium, Cr ⁶⁺) (For non-metal)	mg/kg	With reference to IEC 62321-7-2 Edition 1.0 : 2017, by alkaline/toluene digestion and determined by UV-VIS Spectrophotometer	8	N.D.
폴리브로민화비페닐 (Polybrominated Biphenyls, PBBs)				
모노브로모비페닐 (MonoBB)	mg/kg	With reference to IEC 62321-6 Edition 1.0 : 2015, by solvent extraction and determined by GC/MS	5	N.D.
다이브로모비페닐 (DiBB)	mg/kg		5	N.D.
트라이브로모비페닐 (TriBB)	mg/kg		5	N.D.
테트라브로모비페닐 (TertaBB)	mg/kg		5	N.D.
펜타브로모비페닐 (PentaBB)	mg/kg		5	N.D.
헥사브로모비페닐 (HexaBB)	mg/kg		5	N.D.
헵타브로모비페닐 (HeptaBB)	mg/kg		5	N.D.
옥타브로모비페닐 (OctaBB)	mg/kg		5	N.D.
노나브로모비페닐 (NonaBB)	mg/kg		5	N.D.
데카브로모비페닐 (DecaBB)	mg/kg		5	N.D.
폴리브로민화디페닐에테르 (Polybrominated Diphenyl Ethers, PBDEs)				
모노브로모디페닐에테르 (MonoBDE)	mg/kg	With reference to IEC 62321-6 Edition 1.0 : 2015, by solvent extraction and determined by GC/MS	5	N.D.
다이브로모디페닐에테르 (DiBDE)	mg/kg		5	N.D.
트라이브로모디페닐에테르 (TriBDE)	mg/kg		5	N.D.
테트라브로모디페닐에테르 (TetraBDE)	mg/kg		5	N.D.
펜타브로모디페닐에테르 (PentaBDE)	mg/kg		5	N.D.
헥사브로모디페닐에테르 (HexaBDE)	mg/kg		5	N.D.
헵타브로모디페닐에테르 (HeptaBDE)	mg/kg		5	N.D.
옥타브로모디페닐에테르 (OctaBDE)	mg/kg		5	N.D.
노나브로모디페닐에테르 (NonaBDE)	mg/kg		5	N.D.
데카브로모디페닐에테르 (DecaBDE)	mg/kg		5	N.D.

Tested by : MB Song, EH Cha, BJ Ha

Notes : mg/kg = ppm = parts per million (함량 표시 : 백만분의 일)
< = Less than (결과 값 이하)
N.D. = Not detected (< MDL, 미검출 - 검출한계 이하)
MDL = Method detection limit (검출한계)

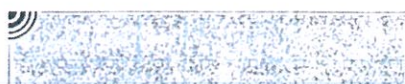


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TEST REPORT (시험 성적서)

발행면수 (PAGE) : 3 of 5

시험성적서-번호 (REPORT NO.) : RT21R-S0694-004-K

발행일자 (DATE) : 2021. 02. 02.

시료 고유번호 (SAMPLE ID NO.) : RT21R-S0694-004

시료명 (SAMPLE DESCRIPTION) : FKM
(FKM)

항목 (TEST ITEM)	단위 (UNIT)	분석방법 (TEST METHOD)	검출한계 (MDL)	시험결과 (RESULT)
헥사브로모시클로데칸 (Hexabromocyclododecane, HBCDD)	mg/kg	With reference to US EPA 3540C, by solvent extraction and determined by GC/MS	10	N.D.

Tested by : BJ Ha

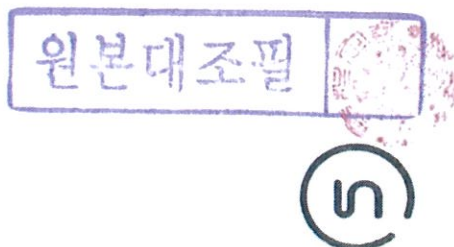
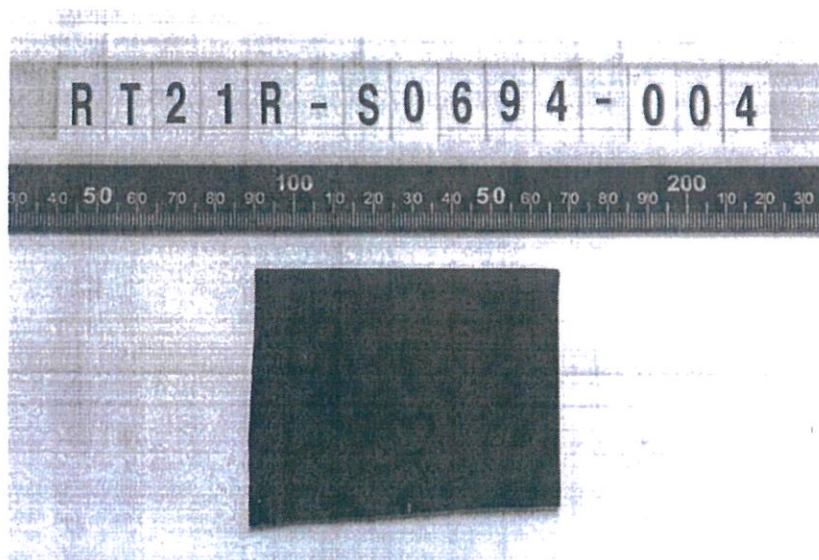
Notes : mg/kg = ppm = parts per million (함량 표시 : 백만분의 일)

< = Less than (결과 값 이하)

N.D. = Not detected (< MDL, 미검출 - 검출한계 이하)

MDL = Method detection limit (검출한계)

* 시료 접수 시 시료 상태 :
(View of sample as received)



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TEST REPORT

(시험 성적서)

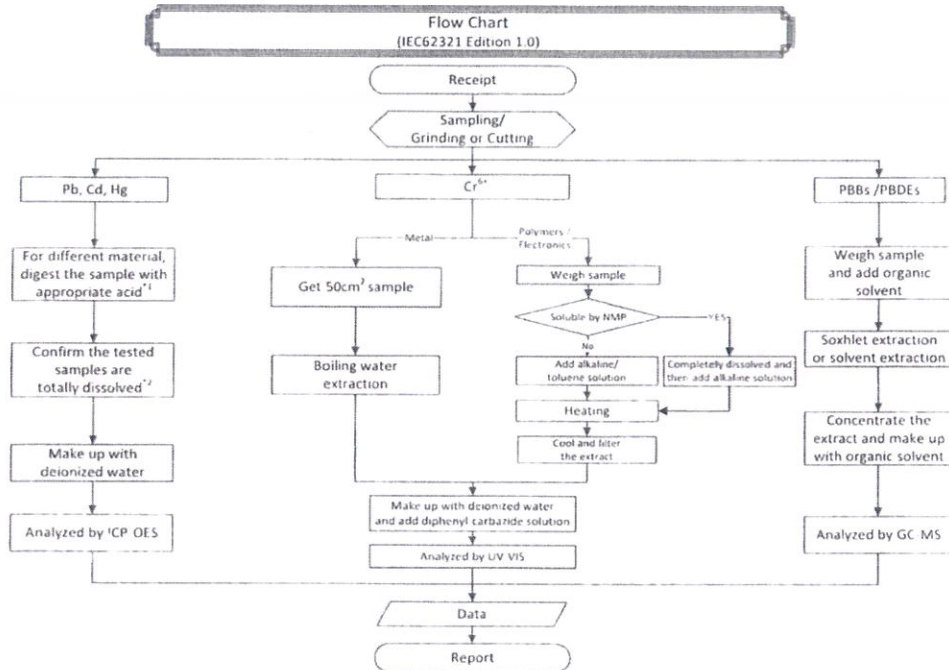
발행면수 (PAGE) : 4 of 5

시험성적서 번호 (REPORT NO.) : RT21R-S0694-004-K

발행일자 (DATE) : 2021. 02. 02.

시료고유번호 (SAMPLE ID NO.) : RT21R-S0694-004

시료명 (SAMPLE DESCRIPTION) : FKM
(FKM)

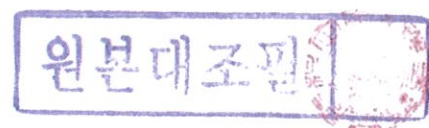


Remarks :

*1 : List of appropriate acid :

Material	Acid added for digestion
Polymers	HNO ₃ , HCl, HF, H ₂ O ₂ , H ₃ BO ₃
Metals	HNO ₃ , HCl, HF
Electronics	HNO ₃ , HCl, H ₂ O ₂ , HBF ₄

*2 : The samples were dissolved totally by pre-conditioning method according to above flow chart.



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발행면수 (PAGE) : 5 of 5

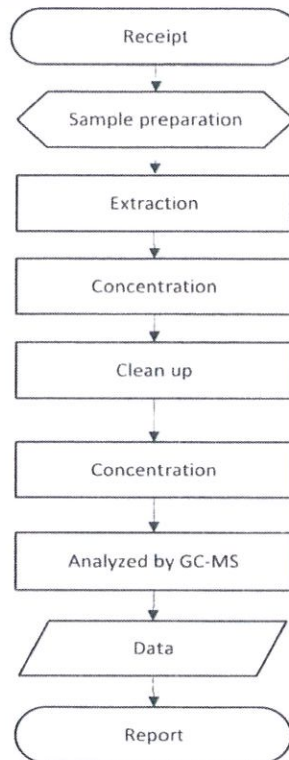
발행일자 (DATE) : 2021. 02. 02.

시험성적서 번호 (REPORT NO.) : RT21R-S0694-004-K

시료 고유번호 (SAMPLE ID NO.) : RT21R-S0694-004

시료명 (SAMPLE DESCRIPTION) : FKM
(FKM)

Flow Chart (HBCDD)



***** End of Report *****

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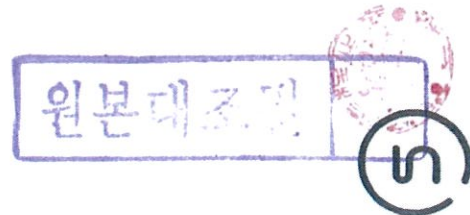
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ASSIST PUMP ASSY- SUB BALL 이탈력 TEST

(주)승정 개발품질팀

차종	품명	품번	시료수	계측기	시험일자	판정
AX	ASSIST-PUMP ASSY-SUB	1058-N90260-10	n=3	PUSH PULL GAUGE	21.07.02	합격

시험조건	-.TEST 온도 : 온도 18℃~22℃ / 습도 70%이하 -.TEST 방법 : BALL 압입(열코킹 전) 완료된 ASSIST PUMP ASSY를 JIG에 고정후 BALL 이탈력 TEST 진행 -.하중방향 : Z축
시험방법	노즐부 절개 후 Ø2.8봉으로 200mm/min 속도로 하중을 가할 것.

평가기준	이탈력은 MIN 11kgf 이상일 것.
------	-----------------------

평가결과





No.	Force	Unit	Displacement	Unit	Judge	Position	Time	Date
1	28.14	kgf	--	--	OK	--	오후 5:52:23	2021-07-02
2	27.08	kgf	--	--	OK	--	오후 6:00:59	2021-07-02
3	26.66	kgf	--	--	OK	--	오후 6:01:53	2021-07-02

소견	TEST 결과 조건만족
----	--------------